

Addressing class imbalance in avalanche forecasting

Manish Kala^{a,*}, Shweta Jain^b, Amreek Singh^a, Narayanan Chatapuram Krishnan^c

^a Defence Geo-Informatics Research Establishment (DGRE), DRDO, Chandigarh, India

^b Indian Institute of Technology (IIT), Ropar, Punjab, India

^c Indian Institute of Technology (IIT), Palakkad, Kerala, India

ARTICLE INFO

Keywords:

Statistical avalanche forecasting
Class imbalance
Random Forests
Resampling
Cost-sensitive learning

ABSTRACT

Natural disasters like avalanches and earthquakes are examples of rare events. Predicting such events using supervised classification machine learning models suffers from the class imbalance problem. The number of non-avalanche days exceed the number of avalanche days, and such data distribution skewness interferes with the construction of decision boundaries to support the decision-making procedure. This paper analyses class imbalance from the perspective of avalanche prediction by involving multiple classification approaches, three oversampling and two undersampling techniques, and cost-sensitive approaches. The supervised approaches aimed to predict days with and without avalanches as binary classification. The study was conducted using past 25 seasons of snow and meteorological parameters recorded for two climatologically diverse avalanche prone regions of Indian Himalayas with different levels of class imbalance. The paper also proposes more extensive use of evaluation metrics like balanced accuracy, geometric mean, Probability of Detection (POD) and Peirce Skill Score (PSS) that are pertinent to imbalanced class domains like avalanche forecasting. Extensive empirical experiments and evaluations amply demonstrate that these class balancing techniques lead to significant improvements in the performance of avalanche forecasting models for both regions, albeit with some variations. The POD values improved to 0.83 for Random Forest classifier, 0.65 for Support Vector Machine classifier and 0.75 for Logistic Regression classifier; PSS values also improved to 0.53, 0.47 and 0.5 for Random Forest, Support Vector Machine, and Logistic Regression classifiers, respectively. These findings are complemented by theoretical insights on the proposed solutions to the class imbalance. Our results suggest that the classification based avalanche forecasting models trained using proposed approaches can serve as valuable supplementary decision support tool for avalanche forecasters.

1. Introduction

Avalanches are a constant threat to the army personnel deployed and the civilians inhabiting the snow-bound regions of the Indian Himalayas. The Defence Geo-Informatics Research Establishment (DGRE), erstwhile Snow and Avalanche Study Establishment (SASE), is tasked with ensuring the safe mobility of troops, tourists, and civilians venturing into these treacherous areas. The passive methods for avalanche danger mitigation consisting of the installation of control structures like snow-nets, snow galleries, etc. are economically unviable and not scalable. The active method of avalanche forecasting that strives to predict the time and site of probable avalanche occurrences across the vast and harsh expanse of the Himalayas is more practical and economical. Avalanche forecasting includes detailed data analysis, an understanding of physical rules, and time and space related scaling

issues (McClung, 2000, 2002; Pozdnoukhov et al., 2008; Sharma and Ganju, 2000) and is often represented as an art based on the instinct and knowledge of human forecasters (LaChapelle, 1980).

Statistical approaches to avalanche forecasting like Nearest Neighbours, Artificial Neural Networks, Support Vector Machines, discriminant analysis, decision trees and clustering model the relationship of historical avalanche occurrences and avalanche danger levels in a region with snow-meteorological parameters collected and snow stability tests at a location representative of the weather and snow conditions in the area (Pozdnoukhov et al., 2008). The Nearest Neighbours approach based on the assumption that similar conditions are likely to result in similar events/outcomes (Buser, 1983; Gassner et al., 2001; Purves et al., 2003; Singh and Ganju, 2004; Singh et al., 2005) is fairly popular due to its interpretability and closeness to the inductive process followed by most forecasters (LaChapelle, 1980). However, the approach is highly

* Corresponding author at: Defence Geo-Informatics Research Establishment (DGRE), DRDO, Sector 37 A, Chandigarh 160047, India.

E-mail addresses: manish.kala.dgre@gov.in (M. Kala), shwetajain@iitpr.ac.in (S. Jain), amreek.dgre@gov.in (A. Singh), ckn@iitpkd.ac.in (N.C. Krishnan).

susceptible to overfitting as the number of features increases (McCollister et al., 2003; Pozdnoukhov et al., 2008). Pozdnoukhov et al., 2008 reported comparable performance of the Support Vector Machines and Nearest Neighbours approaches, but the former used ‘untransformed’ features leading to descriptive forecasts, and could handle more features. Non-linear classification approaches like random forests (Hendrikx et al., 2014; Marienthal et al., 2015; Möhle et al., 2014; Pérez-Guillén et al., 2022; Sielenou et al., 2021; Rosenthal et al., 2002) have also been employed for avalanche forecasting; they can also rank the different variables for their contribution to avalanches (Horton et al., 2020).

Class imbalance is a major issue in many real world domains like genomic datasets, sentiment analysis and opinion mining from the social media, rare event prediction. Rare events are events that occur with low frequency but may cause far-reaching impact. Datasets for rare events like natural disasters (e.g. avalanches, earthquakes, floods, tornados), anthropogenic hazards (e.g. credit card fraud, industrial accidents, network intrusion) and diseases (e.g. cancer diagnosis (Krawczyk et al., 2016)) have significantly fewer instances of the rare event compared to non-event cases (Ali et al., 2013; Haixiang et al., 2017). Imbalanced learning approaches, viz., resampling methods, cost-sensitive learning, and hybrid methodologies (He and Garcia, 2009) can help to detect such events. Other approaches (Haixiang et al., 2017) include treating the problem as anomaly detection instead of classification, shifting the decision threshold (from the ‘ideal’ 0.5 to say 0.4 for better classification of minority class) and pre-weighting examples by assigning higher weights to minority class examples.

Trafalis et al., 2014 built imbalanced learning models to forecast tornados. Pham et al. (2023) employed different undersampling and oversampling approaches like Borderline-Synthetic Minority Oversampling Technique (SMOTE), Condensed Nearest Neighbour, etc. to assess earthquake-induced landslide susceptibility. A novel hybrid resampling approach combining Borderline-SMOTE oversampling with undersampling was used to generate minority flood day instances (Wang et al., 2024). Random Forests along with boosting and bagging were used to overcome the imbalance between the number of flooding and non-flooding events (Ekmekcioglu et al., 2022). Snieder et al., 2021 combined data resampling approaches like random undersampling and oversampling, and SMOTE with bagging and boosting to improve the performance of classifiers for minority high flow events in flood warning systems.

Earlier attempts at handling class imbalance in avalanche datasets include considering days with ‘high’ danger levels issued as avalanche days in addition to actual avalanche days while training (Pant and Ganju, 2006). More recently, Pérez-Guillén et al., 2022 achieved accuracy between 72 % and 78 % for predicting avalanche danger levels with Random Forests based on a ‘quality-controlled’ subset of danger level labels which was better than the performance on actual danger levels issued in public forecasts. Pozdnoukhov et al., 2011 generated 25 ‘virtual’ avalanche day data points centered around the normal distributions of the actual feature values associated with avalanche days to mitigate the imbalance before training the Support Vector Machines. Möhle et al. (2014) proposed Balanced Random Forests (BRF) trained on an equal number of avalanche and non-avalanche instances, and Weighted Random Forests (WRF) with class weights of 5 and 1 for avalanche and non-avalanche classes, respectively. Sielenou et al., 2021 combined Random Forests and oversampling to discriminate between three classes of non-avalanche days, days with moderate activity, and days with high activity. The under-represented moderate and high activity avalanche days were ‘copied as many times as needed’ for more balanced datasets then given to random forests.

The present study addresses the research gap related to the class imbalance in avalanche forecasting. The primary objective of this research is to identify and analyse optimal strategies for handling class imbalance in avalanche forecasting models. Specifically, we seek to:

1. Compare the performance of different classification models when applied to imbalanced avalanche datasets.
2. Evaluate the effectiveness of various resampling and cost-sensitive techniques in improving model performance.
3. Provide practical recommendations for the development of classification models for imbalanced datasets in operational avalanche forecasting setting.

To achieve these objectives, we will implement and test multiple classification models and balancing techniques on real-world avalanche datasets; and analyse the results to determine the most effective approaches and validate the findings against existing studies.

This paper discusses class imbalance in avalanche forecasting datasets and how it is a challenge for most traditional classification approaches to forecasting. It is followed by a brief discussion of four competing Machine Learning based avalanche forecasting models in Section 3. Following that is a detailed description of approaches like undersampling, oversampling, and cost-sensitive learning to overcome the class imbalance. Next in Section 4, it discusses the study area and the dataset used in the analysis. The paper analyses the results of the classification models on imbalanced and resampled data in terms of different evaluation metrics and draws conclusions on the potential of these approaches in handling class imbalances in Section 7. It also discusses future research possibilities in terms of other possible ways to mitigate the class imbalance in avalanche forecasting.

2. Data imbalance in avalanche forecasting

Even though rare events like avalanches may be difficult to detect due to their infrequency (Haixiang et al., 2017); still, misclassifying rare events can lead to heavy costs (He and Garcia, 2009; Seiffert et al., 2007). For example, predicting a non-avalanche day for a day with potential avalanche danger may put lives in peril. The aim is to develop classifiers that provide ‘acceptable’ classification results for both avalanche and non-avalanche classes.

2.1. Imbalanced datasets

The datasets within which one or some of the classes have a much larger number of examples than the others are referred to as the imbalanced datasets (Ali et al., 2013). For the imbalanced datasets, the Imbalance Ratio (IR) (Zhu et al., 2020) is defined as:

$$\text{Imbalance Ratio (IR)} = \frac{n}{p} \quad (1)$$

where p and n are the number of records belonging to positive/avalanche class and negative/non-avalanche class, respectively and $p < n$.

2.2. Causes of data imbalance

While the aim of this study is to address the statistical challenges posed by class imbalance, it is also pertinent to discuss some of the reasons for this imbalance. Avalanche occurrence records face significant uncertainty and gaps due to multiple factors, including an over-dependence on visual observations which are not always feasible or may be delayed (Schweizer et al., 2020), as well as the remoteness and the vastness the area. Additionally, small avalanches often go unreported, and the lack of any effective monitoring mechanisms, particularly during storms makes it quite challenging, to observe them or accurately record their release date (van Herwijnen and Schweizer, 2011). Remotely operated infrasound avalanche detection systems are not scalable; further, the detection is limited for wet-snow and small or medium-sized avalanches (Mayer et al., 2020). Satellite-based avalanche detection holds promise (Eckerstorfer et al., 2018) but is

restricted by its spatial resolution and long revisit period of satellite images available in the public domain.

2.3. Challenges for machine learning based avalanche forecasting models

While developing classification models for avalanche forecasting, due to imbalances in the training datasets, the majority non-avalanche class dominates the classification models, and the minority avalanche days have 'little' contribution in the learning of the model. It is quite challenging for traditional classification models to correctly classify avalanche/non-avalanche days due to the following reasons (Haibo and Ma, 2013; Haixiang et al., 2017; Johnson and Khoshgoftaar, 2019):

1. Most classification-based forecasting models that rely on overall performance metrics like prediction accuracy tend to favour the non-avalanche class, since this bias can still result in high accuracy scores.
2. Conventional classifiers perform well with balanced training sets; for imbalanced data they often excel at identifying the majority non-avalanche days while frequently misclassifying the less frequent avalanche days.
3. These models also misidentify rare avalanche days as noise or mistakenly classify noisy data points as avalanche days, given that both are infrequent patterns.

3. Description of ML models, resampling and cost-sensitive approaches and evaluation metrics for imbalanced learning

3.1. Machine learning models for avalanche forecasting

Worldwide, different classification approaches have been employed for the task of avalanche forecasting. In this paper, we have analysed classifiers based on Logistic Regression, Nearest Neighbours, Support Vector Machines and Random Forest for the task of avalanche days' prediction. These approaches were applied both to imbalanced data as well as the datasets resampled using different data balancing techniques. All these models were implemented in Python using the scikit-learn library (Pedregosa et al., 2011), the optimized hyper-parameter values for the classifiers are mentioned in Table A.7 in Appendix 'A'.

3.1.1. Logistic regression (LR) classifier

The logistic regression classifier (Bishop, 2016) models the relationship between multiple snow-meteorological parameters, $x_k = (x_{k,1}, \dots, x_{k,d})$ and the binary target variable with using weights, $w = (w_1, \dots, w_d)$.

$$\hat{y}_k = g\left(\sum_{i=1}^d w_i x_{k,i}\right) \quad (2)$$

where $d = 10$ is the number of dimensions, $0 \leq \hat{y}_k \leq 1$ are predicted probabilities, $k = 1, 2, \dots, m$ and g is the logistic function.

The binary cross-entropy loss (L) quantifies the similarity between the predicted probabilities ($\hat{y}_k$) and true values (y_k) for the m training examples and is minimized through adjustment of the weights in multiple epochs.

$$L = -\frac{1}{m} \sum_{k=1}^m [y_k \log(\hat{y}_k) + (1 - y_k) \log(1 - \hat{y}_k)], \quad (3)$$

where $y_k \in \{0, 1\}$ and $0 \leq \hat{y}_k \leq 1$

3.1.2. Nearest neighbours (NN) classifier

Nearest Neighbours based classification (Buser, 1983; Gassner et al., 2001; Harrington, 2012) assumes that similar snow and meteorological conditions are likely to result in similar outcomes. It builds a 10-dimensional feature space from the training examples labelled as avalanche/non-avalanche (each training example is represented as a unit vector), and the distance between each test data's unit vector and all training

vectors is computed using:

$$distance(x^1, x^2) = \left(\sum_{k=1}^d |x_k^1 - x_k^2|^2\right)^{1/2} \quad (4)$$

where $distance(x^1, x^2)$ is the distance between points x^1 and x^2 , and x_k^1 denotes the value of point x^1 in the k^{th} dimension..

The test data samples are classified through a simple majority vote of n nearest neighbours, i.e., for the test input x assign the class as the most common class label among its n most similar neighbours. Different values of $n \in \{2, 5, 10\}$ were tried, and $n = 10$ was found to result in the best performance through grid cross-validation, and all the snow-meteorological parameters were equally weighted.

3.1.3. Support vector machine (SVM) classifier

The support vector classifier (Cortes and Vapnik, 1995; Pozdnoukhov et al., 2008, 2011), aims at creating the best hyperplane that segregates the feature space into avalanche and non-avalanche days. We implemented the non-linear classifier based on the Radial Basis Function (RBF) kernel.

For the linear SVM classifier, the separating hyperplane is $w \cdot x + b = 0$, where $\cdot$ represents element-wise multiplication, w is slope and b is the intercept s.t.

$$h(x_k) = \begin{cases} +1 & \text{if } w \cdot x_k + b \geq 0 \\ -1 & \text{if } w \cdot x_k + b < 0 \end{cases} \quad (5)$$

The aim of the classifier is to find the optimal hyperplane (i.e., values of w and b) that effectively separates the data (Fig. 1) into two classes using:

$$\min_{w,b} \frac{1}{2} \|w\|^2 \quad \text{subject to } y_k(w \cdot x_k + b) \geq 1, k = 1, 2, \dots, m \quad (6)$$

For real-world datasets like ours, the soft-margin classifier modifies the constraints of the optimisation problem by adding slack variable ζ_k that permits some misclassifications.

$$\min_{w,b,\zeta} \left(\frac{1}{2} \|w\|^2 + C \sum_{i=1}^m \zeta_i \right) \quad \text{subject to } y_k(w \cdot x_k + b) \geq 1 - \zeta_k, \zeta_k \geq 0, k = 1, 2, \dots, m \quad (7)$$

The regularization parameter C determines how much importance we give to ζ_k . Small values of C give more importance to ζ_k leading to wider margin at cost of some misclassifications.

3.1.4. Random forest (RF) classifier

The last classification model that we evaluated was a Random Forest - an ensemble of multiple classification decision trees built using instance-bagging, i.e., on different sub-samples of the training dataset (Bishop, 2016; Breiman, 2001; Möhle et al., 2014; Sielenou et al., 2021). The decision tree splits the data into avalanche and non-avalanche classes by selecting the best feature at each node D that minimises entropy $H(D)$ (and Gini index) after the split.

$$H(D) = -(p_{ava} \log_2 p_{ava} + p_{non-ava} \log_2 p_{non-ava}) \quad (8)$$

where $H(D)$ is the entropy of node D , and $p_{ava} = \frac{N_{ava}}{N_{ava} + N_{non-ava}}$ and $p_{non-ava} = \frac{N_{non-ava}}{N_{ava} + N_{non-ava}}$ are proportion of samples that belong to avalanche class and non-avalanche class respectively for this node D (N_{ava} and $N_{non-ava}$ being the number of avalanche and non-avalanche instances at node D).

Gini Index of the node D that splits based on feature a is:

$$G(D, a) = \frac{N^{left}}{N^{left} + N^{right}} H(D^{left}(a)) + \frac{N^{right}}{N^{left} + N^{right}} H(D^{right}(a)) \quad (9)$$

where N^{left} and N^{right} represent the number of instances in the left and right child nodes, respectively.

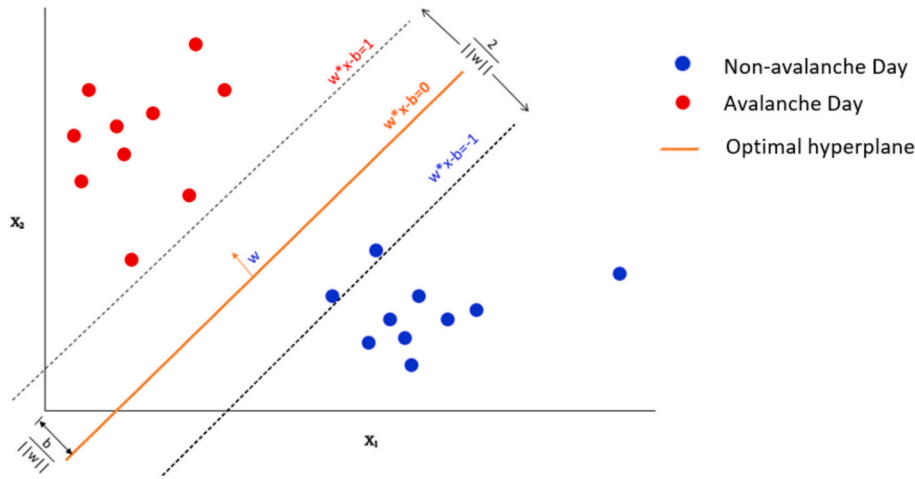


Fig. 1. The optimal hyperplane separating the two classes.

$H(D^{left}(a))$ and $H(D^{right}(a))$ represent the entropy values for the left and right nodes, respectively.

The decision tree strives to perform a split using feature $a^* = \text{argmin}_a G(D, a)$ which results in the minimum Gini Index at each node. The test data point is classified as an avalanche/non-avalanche day by each tree, and the random forest chooses the class based on the majority vote these trees. Since the forest averages the decision, it improves the overall predictive accuracy and is less prone to overfitting. The hyper-parameters, viz., number of trees in the forest, maximum number of features, maximum tree depth, etc. were tuned using grid cross-validation on ‘out-of-bag’ validation samples. The threshold of 0.5 was used for classification, i.e., output values greater than 0.5 were deemed as avalanche days.

3.1.5. Machine learning models and class imbalance

Due to the fact that avalanches are rare and the number of non-avalanche days outnumber the avalanche days by a wide margin, the above-mentioned classifiers fail to give desirable performance for the avalanche days. Table 1 mentions the issues due to the class imbalance with these classifiers.

Table 1

Avalanche forecasting approaches and their issues due to class imbalance.

Classification approach to avalanche forecasting	Issues due to class imbalance
Logistic Regression Classifier	a) Loss function is averaged over all training data points, so the majority non-avalanche class dominates it; the avalanche class examples don't participate enough. b) The classifiers fail to learn the underlying distributive characteristics of the avalanche class.
Nearest Neighbours Classifier	Most of the n nearest neighbours of the avalanche class query point may belong to the non-avalanche class (due to its higher density in feature space), resulting in the misclassification of avalanche days as non-avalanches and thus more false negatives for the avalanche day instances.
Support Vector Machine Classifier	A higher density of non-avalanche days near the class boundary region skews the separating hyperplane towards minority avalanche class to reduce the total number of misclassifications; this results in more avalanche days being predicted as non-avalanche days.
Random Forest Classifier	The decision trees of the forest constructed from imbalanced data using bootstrapped sampling may contain few avalanche day instances, resulting in a skewed ‘Gini Index’, and thus the trees give poor performance for predicting avalanches.

3.2. Approaches to handle imbalanced learning in avalanche datasets

Resampling techniques (Table 2) mitigate the negative effects of skewness in the class distribution of data on the learning model by ‘rebalancing’ the sample space. The major advantages of the resampling techniques are that they are independent of the classification model to be employed subsequently and are easier to implement (Haixiang et al., 2017; He and Garcia, 2009). The resampled dataset need not have an exactly balanced distribution but instead a distribution that can be handled by the classification-based avalanche forecasting models.

3.2.1. Undersampling methods

These methods discard instances of the majority class, i.e., non-

Table 2

Different approaches to mitigate class imbalance and their types.

Type	Approach	Abbreviation	Description
Undersampling	Random Undersampling	RUS	Randomly remove majority class instances
	NearMiss Undersampling	NMUS	Selectively remove majority class instances closest to minority class instances
Oversampling	Random Oversampling	ROS	Randomly duplicate minority class instances
	Synthetic Minority Oversampling Technique	SOS	Generate synthetic minority class instances through interpolation between nearest neighbours of minority class
	KMeansSMOTE Oversampling	KSOS	Cluster minority class samples into subgroups and then apply SMOTE to suitable subgroups for synthetic instance generation
Cost-sensitive learning	Cost-sensitive Logistic Regression classifier	CS_LR	Include different costs for misclassifying majority and minority class instances
	Cost-sensitive Support Vector Machine classifier	CS_SVM	Modify optimization objective to include class-specific weights
	Cost-sensitive Random Forest classifier	CS_RF	Include different weights for majority and minority classes while calculating the entropy of the chosen split point

avalanche days, to achieve some balance in the data distribution (Osorio et al., 2021).

3.2.1.1. Random undersampling (RUS). The approach randomly removes the records belonging to non-avalanche days (Liu et al., 2008). It reduces the runtimes of classification models by reducing the number of training examples; however, the removal of large amounts of training data can significantly reduce the total amount of information the model can learn from. In our experiments, we randomly undersampled the non-avalanche days so that the imbalance ratio was reduced to multiple ratios of 5:1, 4:1, 3:1, and 2:1 for extensive empirical studies.

3.2.1.2. NearMiss undersampling (NMUS). Keeping in mind the drawbacks of random undersampling, we tried NearMiss (Mani and Zhang, 2003) undersampling, which involves:

- For all points belonging to the non-avalanche day calculate their Euclidean distance from all the avalanche day points.
- Remove the non-avalanche day instances for which the average distance to minority avalanche day instances are the smallest; the aim being to increase the spaces between the two classes.

Unlike random undersampling, this approach ensures that the data is more evenly distributed and the models are less biased.

3.2.2. Oversampling methods

These methods create new instances in the avalanche days' class space. In the present study, we have employed three oversampling methods, viz., random duplication of minority avalanche day instances, SMOTE, and KMeansSMOTE.

3.2.2.1. Random oversampling (ROS). In Random Oversampling, the data instances for avalanche days are repeatedly copied in the original dataset until a more balanced data distribution is achieved. Unlike undersampling, it does not lead to any information loss. However, this duplication only increases the number of examples without providing any new information about the avalanche class. Further, it may result in overfitting and poorer generalization performance (He and Garcia, 2009).

3.2.2.2. Synthetic minority oversampling technique (SOS). The SMOTE algorithm (Chawla et al., 2002; Hosseinzadeh and Eftekhari, 2015) generates 'synthetic' instances of avalanche days based on the similarities between existing minority instances in the feature space. For creating a new synthetic avalanche day instance ($\vec{a}$), an instance of the avalanche day class (say $\vec{a}$) is selected and its n nearest avalanche day

class neighbours are found (Fig. 2). $\vec{a}$ is produced by randomly choosing one of the n nearest neighbours (say $\vec{b}$) and forming a vector $\vec{a}$ between $\vec{a}$ and $\vec{b}$ in the feature space, mathematically $\vec{a} = \vec{a} + w \times (\vec{b} - \vec{a})$, where w is a random weight in $[0,1]$.

SMOTE mitigates overfitting resulting from random oversampling (to a certain extent) since it does not simply replicate the avalanche days (Sielenou et al., 2021) but synthetically generates new instances. Moreover, since the new synthetic instances are comparatively closer to already existing minority examples in the feature space, they are more credible. The pseudocode and Python implementation of SMOTE can be found at our GitHub repository: <https://github.com/ManishKalaPHD/CI-assImbalance>.

3.2.2.3. KMeansSMOTE oversampling (KSOS). There are certain limitations with SMOTE that restrict the generation of optimal synthetic avalanche class instances (Douzas et al., 2018; Maldonado et al., 2019).

- By randomly selecting instances from the minority avalanche class, SMOTE might overlook adjacent instances from non-avalanche class when creating synthetic instances, resulting in overlapping between the classes.
- While regions with fewer avalanche instances might remain under-represented, areas with higher concentration of minority avalanche instances are more prone to excessive inflation.
- SMOTE could amplify noise by interpolating a 'noisy' avalanche instance situated among non-avalanche instances.

A new approach, KMeansSMOTE (Douzas et al., 2018) overcomes these shortcomings of SMOTE and includes the following steps:

- Clustering:** Cluster the entire input space into k clusters using k -means clustering.
- Selection:** Select clusters with a high number of minority avalanche day instances (to avoid generating synthetic avalanche days in overlapping areas).
- SMOTEing:** Generate synthetic avalanche data instances in inverse proportion to the density of the avalanche day instances in the 'selected' clusters, i.e., generate more synthetic instances in the clusters with a sparser population of minority avalanche day instances.

In this oversampling approach, k is a hyper-parameter that needs tuning since it controls the number of clusters generated. Different values in the range of 2 to 10 were tried, and the value of $k = 5$ was found to give better results for the data considered in the study. The pseudocode and Python implementation of KMeansSMOTE can be found

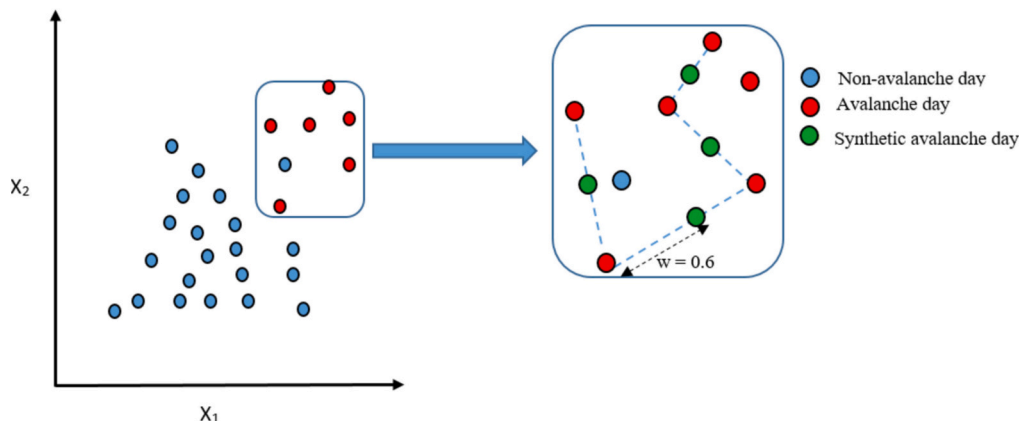


Fig. 2. Generation of 'synthetic' avalanche day class data points using SMOTE.

at our GitHub repository: <https://github.com/ManishKalaPHD/ClassImbalance>.

3.3. Cost-sensitive learning

Most ML based classification approaches assume equal misclassification errors, but often missing an avalanche day is far worse than incorrectly classifying a non-avalanche day. Cost-sensitive learning strives for explicitly defining and using different costs when training the models (Moepya et al., 2014) through the concept of classification cost:

$$\text{Classification Cost (CC)} = C(0, 0) \times \text{TN} + C(1, 1) \times \text{TP} + C(0, 1) \times \text{FP} + C(1, 0) \times \text{FN} \quad (10)$$

where, TN, TP, FP, FN are the number of true negatives, true positives, false positives, and false negatives (defined in Section 3.4).

$C(0, 0)$ and $C(1, 1)$ are costs of correctly classifying non-avalanche and avalanche days, respectively.

and $C(0, 1)$ and $C(1, 0)$ are costs of misclassifying non-avalanche and avalanche days, respectively.

Generally $C(0, 0) = C(1, 1) = 0$; no associated costs if correctly classified.

and also $C(1, 0) > C(0, 1)$; $C(1, 0) \neq 0$, $C(0, 1) \neq 0$ misclassifying an avalanche day as a non-avalanche day is costlier than otherwise.

The Non-Cost Sensitive (NCS) models, treat all misclassification errors equally without prioritizing one class over another, for them $C(0, 1) = C(1, 0)$. The costs are often supplied by domain experts, and their values determine the quality of the models. Möhle et al., 2014 used class weights of 5 and 1 for avalanche and non-avalanche classes, respectively, i.e. $C(1, 0) = 5$ and $C(0, 1) = 1$. Sometimes the misclassification costs are assigned as inverse class distribution ratios; for the imbalance ratio of 1:9.5, we may assume $C(1, 0) = 9.5$ and $C(0, 1) = 1$. In the present study, we evaluated cost-sensitive versions of the Logistic Regression, Support Vector Machine, and Random Forest classifiers for different cost ratios like inverse class ratio cost, 4:1 and 2:1.

3.3.1. Cost-sensitive logistic regression classifier

The negative log likelihood cost function (L), is modified to include the different misclassification costs to skew the decision boundary towards minority avalanche class instances to reduce the total misclassification cost.

$$L = -\frac{1}{m} \sum_{k=1}^m C(1, 0) (y_k \log(\hat{y}_k)) + C(0, 1) ((1 - y_k) \log(1 - \hat{y}_k)) \quad (11)$$

3.3.2. Cost-sensitive support vector machine classifier

The regularization parameter (C) for soft margin Support Vector Machine classifier controls the trade-off between minimization of class misclassification and maximisation of separation margin; it thus permits some points to appear on the incorrect side of the decision boundary. In traditional classification, since each class has the same weight, the soft margin is symmetrical leading to the favouring of the dominant non-avalanche class (Batuwita and Palade, 2013). Now in the cost-sensitive Support Vector Machine classifier, we weigh the margin parameter C in proportion to the class importance (Veropoulos et al., 1999).

$$C_{non-ava} = \frac{C}{w_{non-ava}} \text{ and } C_{ava} = \frac{C}{w_{ava}} \quad (12)$$

where $C_{non-ava}$ and C_{ava} are the soft margin parameters for the non-avalanche and avalanche classes, respectively. And $w_{non-ava}$, w_{ava} are importance/weight for non-avalanche and avalanche classes, respectively.

Normally, $w_{non-ava} < w_{ava}$ (and thus $C_{non-ava} > C_{ava}$), larger C values for non-avalanche instances result in smaller penalties for their

misclassification because of a softer margin. Now the modified optimization problem becomes:

$$\min_{w, b, \xi} \left(\frac{1}{2} \|w\|^2 + C_{ava} \sum_{(i|y_i=+1)} \xi_i + C_{non-ava} \sum_{(i|y_i=-1)} \xi_i \right) \quad (13)$$

subject to $y_i(w \cdot x_i + b) \geq 1 - \xi_i$, $\xi_i \geq 0$, $i = 1, 2, \dots, m$

where $y_i = +1$ is the avalanche class and $y_i = -1$ is the non-avalanche class.

As shown in Fig. 3, this arrangement encourages the separating margin to skew away from the avalanche class instances and thus be more accommodative to misclassification of non-avalanche days into avalanche days.

3.3.3. Cost-sensitive Random Forest classifier

For a cost-sensitive Random Forest classifier, we include the weight/contribution of the avalanche and non-avalanche classes while calculating the entropy $H(D)$ of the chosen split point D . So instead of taking the proportion of samples that belong to avalanche and non-avalanche classes p_{ava} and $p_{non-ava}$ as discussed in Section 3.1.4, we use p'_{ava} and $p'_{non-ava}$ while calculating the entropy and Gini Index.

$$p'_{ava} = \frac{w_{ava} N_{ava}}{w_{ava} N_{ava} + w_{non-ava} N_{non-ava}} \text{ and } p'_{non-ava} = \frac{w_{non-ava} N_{non-ava}}{w_{ava} N_{ava} + w_{non-ava} N_{non-ava}} \quad (14)$$

where $w_{ava} > w_{non-ava}$, $w_{ava} \neq 0$ and $w_{non-ava} \neq 0$

The use of higher weights for the avalanche class places a heavier penalty on its misclassification by the Random Forest classifier and overcomes the bias of the traditional Random Forest approach towards the majority non-avalanche class instances.

3.4. Evaluation metrics

For evaluating the performance of classifiers, we normally make use of a confusion matrix (Wilks, 2011). The confusion matrix (Table 3) has four values of interest:

True Positives (TP) = number of actual avalanche days that have been correctly classified as avalanche days.

False Positives (FP) = number of actual non-avalanche days that have been incorrectly classified as avalanche days.

False Negatives (FN) = number of actual avalanche days that have been incorrectly classified as non-avalanche days.

True Negatives (TN) = number of actual non-avalanche days that have been correctly classified as non-avalanche days.

Using the confusion matrix, several standard evaluation metrics (Table 4) may be defined. Accuracy measures the number of correct predictions over the total number of prediction but it is a highly inappropriate evaluation metric since it is influenced by the majority class of non-avalanche days. For class imbalance, where 90 % of the dataset is made up of non-avalanche days; it is trivially possible to achieve 90 % (or 0.90) accuracy if one simply predicts all days as non-avalanche days. This high level of accuracy is oblivious to the classifiers' poor performance on the avalanche class, the class that matters more.

Probability of Detection or True Positive Rate measures the proportion of the actual avalanche days that were correctly predicted to be avalanche days by the classifier, a smaller number of false negatives results in better Probability of Detection values. While the aim is to improve the performance of classifiers for the avalanche class, it should not come at a heavy cost to the non-avalanche class, so suitable metrics are required to evaluate the performance on the non-avalanche class too. True Negative Rate refers to the ratio of correctly classified non-avalanche days to the total number of non-avalanche days in the data, i.e., it tells how well did our models catch the non-avalanche cases.

Balanced Accuracy (Brodersen et al., 2010) combines True Negative

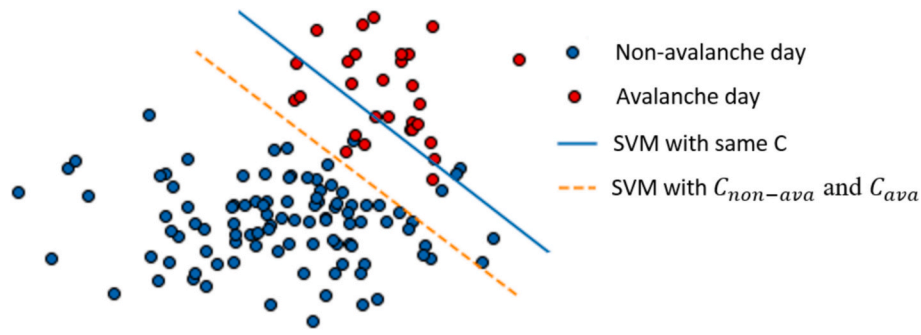


Fig. 3. Hyperplanes for Support Vector Machine (SVM) classifiers with the same C value for both classes and different C values for avalanche and non-avalanche classes.

Table 3
Confusion matrix for binary classification.

		Observed	
		Avalanche Day	Non-avalanche Day
Predicted	Avalanche Day	True Positive (TP)	False Positive (FP)
	Non-avalanche Day	False Negative (FN)	True Negative (TN)

Rate and True Positive Rate values to compute a metric that is more sensitive to the minority avalanche class. It overcomes the limitations of accuracy by giving equal importance to the relative proportions of avalanche day and non-avalanche day instances. In a scenario with 90 % non-avalanche days, even if we predict all days as non-avalanche days ($TP = 0$, $FP = 0$), Balanced Accuracy is only 0.5, unlike Accuracy which is 0.9. Geometric Mean combines both the True Negative Rate and True Positive Rate metrics (Seiffert et al., 2007), thus considering the performance on both avalanche and non-avalanche instances. Since $\frac{x+y}{2} \geq \sqrt{xy}$, so $\text{Balanced Accuracy} \geq \text{Geometric Mean}$; difference between Geometric Mean and Balanced Accuracy increases when the difference between True Positive Rate and True Negative Rate increases. Geometric Mean is more affected by lower values, $\text{Geometric Mean} = 0$ when any of the two classes has zero accuracy (Vuttipittayamongkol et al., 2021). So Geometric Mean is reflective of such unfavourable scenarios, while Balanced Accuracy only provides average values in such cases.

Heidke Skill Score (Ebert and Milne, 2022) takes into account the imbalance by evaluating the classifiers' skill relative to a no-skill or random forecast model, it considers both correct and incorrect predictions across avalanche and non-avalanche classes. The random guess baseline model in the Heidke Skill Score, is typically adjusted to reflect the observed class frequencies, rather than assuming equal probability for each class. Peirce Skill Score (Ebert and Milne, 2022) measures how

well the classifier discriminates between the occurrence and non-occurrence of rare events, it is particularly useful since it accounts for both false alarms and missed events.

4. Study area and data preparation

4.1. Study area

The two climatologically diverse areas selected for the present research are the Chowkibal-Tangdhar (CT) region in Pir Panjal range and Drass-Kargil (DK) region in Great Himalayan range of the Western Himalayas in Jammu and Kashmir, Union Territory, India. While the CT region may be classified to have maritime climate, the DK region has characteristics of continental climate (Haegeli and McClung, 2007; Singh et al., 2015). The CT region has a 36.18 km road that crosses the Pir Panjal range at Nasta-Chun (NC) pass (Joshi and Srivastava, 2014; Singh et al., 2005) with 25 registered avalanche paths (Fig. 4). The elevation of potential avalanche release areas of avalanche paths in the area ranges from about 2350 m to 3600 m above msl. In DK region the roads are affected by 17 major avalanche paths.

4.2. Snow and meteorological parameters

During the Indian winter season, the region records moderate to nearly heavy snowfall due to western disturbances - the extratropical storms originating in the Mediterranean. The snow and meteorological data collected at Neelam Stage-II observatory (latitude $34^{\circ} 24' 57''$, longitude $73^{\circ} 59' 5''$, altitude 2650 m) are assumed to represent the general weather conditions in the CT region. The Drass observatory (latitude $34^{\circ} 25' 34''$, longitude $75^{\circ} 46' 20''$, altitude 3250 m) is used to represent DK region. For operational avalanche forecasting, the forecasters perform critical analysis of meteorological parameters and snowpack properties influencing snowpack stability (Möhle et al.,

Table 4
Evaluation Metrics for imbalanced datasets used in the study.

Metric	Abbreviation	Formula	Definition	Range
Accuracy	ACC	$\frac{(TP + TN)}{(TP + FP + TN + FN)}$	Number of correct predictions over total number of predictions for both avalanche and non-avalanche days	[0,1]
True Positive Rate or Probability of Detection	POD	$\frac{TP}{TP + FN}$	Proportion of actual avalanche days that were correctly predicted to be avalanche days	[0,1]
True Negative Rate	TNR	$\frac{TN}{TN + FP}$	Proportion of actual non-avalanche days correctly predicted to be non-avalanche days	[0,1]
Balanced Accuracy	BA	$\frac{TPR + TNR}{2} = \frac{TP}{2(TP + FN)} + \frac{TN}{2(TN + FP)}$	Average of TPR and TNR	[0,1]
Geometric Mean	GM	$\sqrt{TPR \times TNR}$	Considers relative performance on both avalanche and non-avalanche days	[0,1]
Heidke Skill Score	HSS	$\frac{2(TP \times TN - FP \times FN)}{(TP + FP)(FP + TN) + (TP + FN)(FN + TN)}$	Measures fractional improvement of the forecast over random forecast	$[-\infty, 1]$
Peirce Skill Score or True Skill Statistic	PSS	$\frac{TP \times TN - FP \times FN}{(TP + FN)(FP + TN)}$	Measures ability to discriminate between avalanche and non-avalanche days	$[-\infty, 1]$

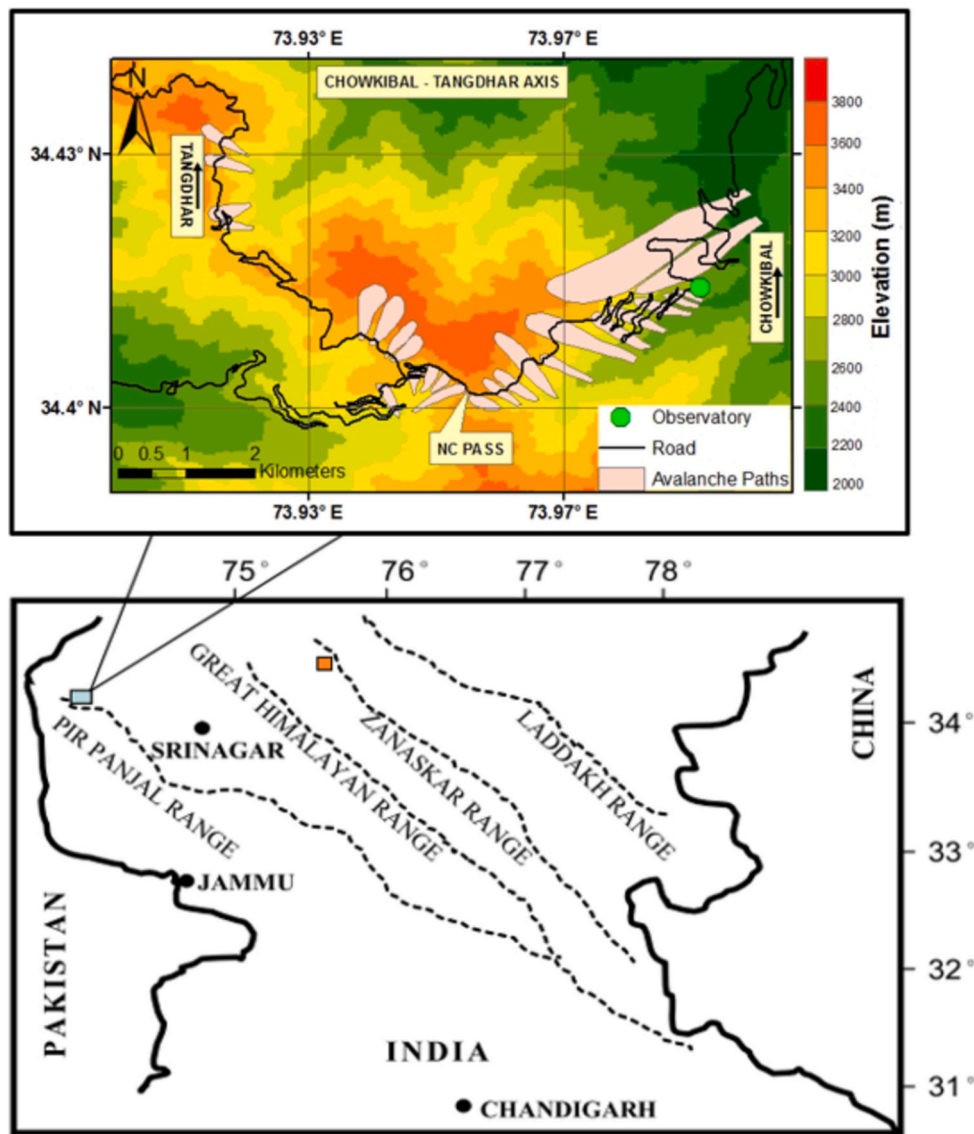


Fig. 4. Chowkibal-Tangdhar (CT) region with Neelam Stage-II observatory, road axis, and avalanche paths. CT and DK regions shown in cyan and orange rectangles, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 5
Snow and meteorological parameters for CT axis.

Parameter	Description	Time of record	Unit	Range	Mean Value
Air temperature	Dry bulb temperature recorded	0830 h (d)	$^{\circ}\text{C}$	[−14,16.5]	0.38
Air temperature change in last 24 h	Difference between air temperatures of d and d-1 day	Air temperature change between 0830 h (d) and 0830 h (d-1)	$^{\circ}\text{C}$	[−12.7,13.2]	0.05
Snow surface temperature	Snow surface temperature recorded	0830 h (d)	$^{\circ}\text{C}$	[−21,0]	−3.12
Wind speed	Average wind speed for last 24 h	0830 h (d)	m/s	[0,8.35]	1.23
Height of new snow in last 1 day	Depth of freshly fallen snow accumulated on snow board during last 1 day (snowboard reset daily at 0830)	Fresh snowfall from 0830 h (d-1) to 0830 h (d)	cm	[0,130]	5.38
Height of new snow in last 2 days	Depth of freshly fallen snow accumulated on snow board during last 2 days	Height of new snow in last 1 day + new snow between 0830 h (d-2) to 0830 h (d-1)	cm	[0,226]	10.76
Height of new snow in last 3 days	Depth of freshly fallen snow accumulated on snow board during last 3 days	Height of new snow in last 2 days + new snow from 0830 h (d-3) to 0830 h (d-2)	cm	[0,306]	16.14
Height of snowpack	Vertical distance from base to snow surface	0830 h (d)	cm	[0,332]	68.91
Snow water equivalent	Water equivalent of total seasonal snow recorded	0830 h (d)	mm	[0,1543.4]	420.75
Penetration of snow surface	Penetration depth of first element of rammsonde (1 m, 10 N)	0830 h (d)	cm	[0,99]	9.89

2014). The objective is to predict potential avalanche activities in the region based on the statistical correlation between past avalanche occurrence events in the region and the snow-meteorological parameters of the observatory and to apply it in an operational forecasting setting.

The following ten parameters (also considered by Gassner et al., 2001; Singh et al., 2005; Singh et al., 2015 and Möhle et al., 2014) have been considered for avalanche forecasting (Table 5). These parameters were taken into consideration due to the fact that they are being generated (directly or derived) using the WRF model (Skamarock et al., 2021) and are used for operational avalanche forecasting at DGRE. The parameters are believed to be critically connected with avalanche occurrences as suggested in previous studies. Height of snowpack, snow water equivalent of the season and penetration of snow surface are relevant to ascertain the amount of snow load and snow stability on the avalanche prone slopes. Snowfall over the last 3 days gives a fair idea about the increase in the snow load over a period of time; wind speed represents the likelihood of cornices formation in the region due to snow drift. Snow surface temperature, air temperature and its change over 24 h influence the stability of snowpack by affecting bonding between layers, rate and type of metamorphism. Recent studies (Pérez-Guillén et al., 2022; Viallon-Galinier et al., 2023) suggest that advanced snow stability variables result in better predictive performance. However, we have taken these parameters since they can be easily measured across the semi-automated and manual observatories spread over the Indian Himalayas.

- Daily meteorological parameters:* Air temperature, wind speed during last 24 h, height of new snow in last 1 day.
- Parameters for recent meteorological conditions:* Air temperature change in 24 h, height of new snow in last 2 days and 3 days.
- Snowpack parameters:* Snow surface temperature, penetration of snow surface, snow water equivalent of the season, height of snowpack.

While the three daily meteorological parameters i.e. snow surface temperature, penetration of snow surface, and height of the snowpack are recorded directly at 0830 h at the observatories; the three parameters for recent meteorological conditions and snow water equivalent of the season are derived from past data. Further, while air temperature and wind speed are recorded through automated sensors, all other parameters are manually recorded. The summary statistics of the parameters recorded for the Neelam Stage-II observatory during the 25 winter seasons (November–April) between 1997 and 2022 are in Table 5 (where d is the day under consideration, d-1 is the previous day, and d-2 is the pre-previous day).

4.3. Data preparation

In the present study, records of 4531 days (approximately 181 days in each season) containing the values for 10 snow-meteorological parameters along with the associated avalanche occurrences/non-occurrences on those days during the winter season (November–April) of 25 years (1997–2022) were analysed. The avalanche days used in the study are those with one or more avalanches visually observed and reported to DGRE by army personnel or civilians venturing in the regions. The road axes in these regions encounter heavy vehicular and pedestrian traffic even during the winter periods, and thus the avalanche occurrence are generally reported. However, there may still be some instances of non-reporting of avalanches particularly those not visible from roads and occurring during storms. Despite these limitations, DGRE has been maintaining quite comprehensive record of avalanche activities in the regions.

We did not discriminate between the avalanche paths, and days with one or more avalanche events (both dry- and wet-avalanches) in any of the avalanche paths were considered avalanche days for respective

regions. As a result, the magnitude of daily avalanche activity becomes irrelevant, and errors in avalanche data are limited to the issues of timing and misinterpretation of avalanche types. Records with missing values for one or more parameters were discarded, and the remaining records were used for further processing. There were just 407 avalanche days (Table 6a) among the total 4279 days for which the records were analysed in the CT region. For the DK region, there were 290 avalanche days out of the 4167 days processed (Table 6b).

For the CT region the imbalance ratio is around 9.5:1, i.e., for each avalanche day there are around 9.5 non-avalanche days in the corresponding period. The scale at which we observe avalanche occurrences can significantly affect the balance of data, larger areas are more likely to witness at least one avalanche on any given day. Since the DK region covers a much smaller area, the imbalance is around 13.3 during the same period. The data was divided into training and test sets in a 80:20 ratio by randomly drawing individual days using stratified sampling maintaining the same class distribution in the training and test sets. This splitting into train and test datasets by individual days (and not by grouping days by seasons), reduces the risk of overfitting to particular time periods and enhances the models' capacity to generalize to new and unseen data. The snow-meteorological parameters were min-max normalised in the range [0,1] so that larger-valued parameters like height of snowpack do not dominate the smaller-valued parameters during the learning phase of the supervised classification. The class labels were encoded such that '1' represented the avalanche day and '0' described the non-avalanche day; the classification thus became a binary class classification.

The training dataset was further divided into train and validation sets using 5-fold cross validation, the different models were trained five times, each time using 4 folds for training and the remaining 1 fold as the holdout validation set. The use of five-fold cross validation while training ensured that the results were generalised and the classification models were not prone to overfitting. Now the training data (and not the validation and test sets) was resampled through different approaches of undersampling and oversampling to train the classifiers and evaluate their performance on the test sets. By resampling only the training data, we aimed to train the models effectively from the available data without compromising the integrity of the test set, which remained 'untouched' for unbiased evaluation. This approach ensures that the models are evaluated on their ability to generalize to unseen and imbalanced avalanche datasets.

5. Results

5.1. Performance on imbalanced data

We started by applying all classification methods listed in Section 3.1 without resampling instances in the avalanche and non-avalanche classes. The performance of the trained models was then evaluated

Table 6

Number of avalanche and non-avalanche days in the train and test dataset for (a) CT region and (b) DK region.

Information	Overall dataset	Train dataset	Test dataset
(a) CT Region			
Percentage	100 %	80 %	20 %
Total Records	4279	3423	856
Number of Avalanche Days	407	325	82
Number of Non-Avalanche Days	3872	3098	774
Imbalance Ratio (IR)	9.5:1	9.5:1	9.5:1
(b) DK Region			
Percentage	100 %	80 %	20 %
Total Records	4167	3334	833
Number of Avalanche Days	290	232	58
Number of Non-Avalanche Days	3877	3102	775
Imbalance Ratio (IR)	13.3:1	13.3:1	13.3:1

Table 7

Performance of different classifiers trained with original imbalanced data on test set of a) CT region and b) DK region.

Classifier	Evaluation Metrics for CT						
	ACC	POD	TNR	BA	GM	HSS	PSS
(a)							
LR	0.86	0.31	0.94	0.62	0.54	0.29	0.25
NN	0.84	0.41	0.90	0.66	0.61	0.30	0.31
SVM	0.87	0.15	0.98	0.57	0.38	0.16	0.13
RF	0.85	0.51	0.90	0.70	0.68	0.38	0.41
(b)							
LR	0.85	0.18	0.98	0.58	0.42	0.23	0.16
NN	0.86	0.28	0.97	0.63	0.52	0.33	0.26
SVM	0.84	0.17	0.97	0.57	0.41	0.19	0.14
RF	0.85	0.33	0.95	0.64	0.56	0.34	0.28

using the evaluation metrics mentioned in Section 3.4 on the test dataset for both the regions (Table 7).

(b)

5.2. Performance on undersampled non-avalanche days

The random and NearMiss undersampling schemes discussed in Section 3.2.2 were applied, and the resampled datasets were used as training inputs for the classification models and the performance was evaluated on the test datasets. Fig. 5 shows the performance of the Nearest Neighbours and Random Forest classifiers trained using imbalanced dataset and the undersampled datasets on the test set for CT region. The full results in tabular form are included in Tables A.1 and A.2 in Appendix A.

5.3. Performance on oversampling avalanche day instances

Three oversampling techniques viz. random, SMOTE, and KMeansSMOTE were attempted in the present study for imbalance ratios of 2:1, 3:1, 4:1, and 5:1. The comparable performance of the oversampling approaches in terms of True Negative Rate, Probability of Detection, Geometric Mean, and Peirce Skill Score for both CT and DK regions is shown in Fig. 6. The full results in tabular form are in Tables A.3 and A.4 in Appendix A.

5.4. Performance of cost-sensitive classifiers

Finally, the cost-sensitive approaches of Section 3.3 were applied and evaluated for performance. The cost-sensitive versions of Logistic Regression, Support Vector Machine, and Random Forest were attempted in the present study. In addition to the inverse class ratio cost (IR), we tried these classifiers with a range of other costs too i.e. 4:1 and 2:1 (where cost 2:1 means the cost of misclassifying an avalanche day as non-avalanche $C(1, 0) = 2$ while the cost of misclassification of a non-avalanche day $C(0, 1) = 1$). Fig. 7 shows the relative performance of

Logistic Regression and Support Vector Machine based cost-sensitive classifiers for different cost ratios. The full results in tabular form for CT and DK regions are in Tables A.5 and A.6 respectively in Appendix A.

6. Discussion

6.1. Performance on imbalanced data

As expected, no resampling still results in high overall accuracy levels (Table 7) for almost all classifiers (>84 % for all 4 classifiers) for both regions; but since the data is imbalanced and we are more interested in detecting instances of avalanche days, this evaluation metric is not the most appropriate. Despite high accuracy values, most of the classifiers did not fare well in other evaluation metrics that consider performance on both classes. For CT region, all classifiers (except Random Forest) fared poorly in detecting avalanche days (Probability of Detection <0.4); even the Heidke Skill Score and Peirce Skill Score values were less than 0.3 for them except for Random Forest. Similarly, for the DK region, Probability of Detection was less than 0.35 for all the classifiers, Balanced Accuracy and Geometric Mean values were also not satisfactory. For both the regions, the Random Forest classifier performed comparatively better than others in terms of Probability of Detection, Geometric Mean, Heidke Skill Score, and Peirce Skill Score. A possible explanation for the better performance is that it is an ensemble of multiple classification decision trees built using instance-bagging, so it is less prone to the issues of class imbalance. As expected, the True Negative Rate values were very high (>0.9) for both the regions since the classifiers were performing well at detecting non-avalanche days. Almost all the classifiers performed better for the CT region as compared to DK region.

6.2. Performance on undersampled non-avalanche days

In CT region, for the imbalance ratio of 5:1 (Table A.1), while the True Negative Rate did not deteriorate much for these classifiers, there was some improvement in other metrics for almost all classifiers for both RUS and NMUS, though the latter performed better. The biggest gainers were Logistic Regression and Support Vector Machine classifiers. For more balanced ratio of 2:1 (Table A.2), there was further improvement in Probability of Detection criteria for all classifiers; with NearMiss Undersampling it improved from 0.51 to 0.83 for Random Forest, from 0.15 to 0.65 for Support Vector Machine, and from 0.31 to 0.75 for Logistic Regression; the Balanced Accuracy and Geometric Mean values improved to around 0.70, signifying improved performance on avalanche days class; the accuracy reduced between 10 % and 20 %, but it was still not less than 0.7 for all classifiers, which is acceptable given the improvement in Probability of Detection, Balanced Accuracy, and Geometric Mean. Peirce Skill Score values with NearMiss Undersampling also improved to 0.5 (from 0.25), 0.47 (from 0.31), 0.47 (from 0.13.) and 0.53 (from 0.41) for Logistic Regression, Nearest Neighbours, Support Vector Machine, and Random Forest classifiers, respectively (Fig. 5). As expected, the True Negative Rate values reported some

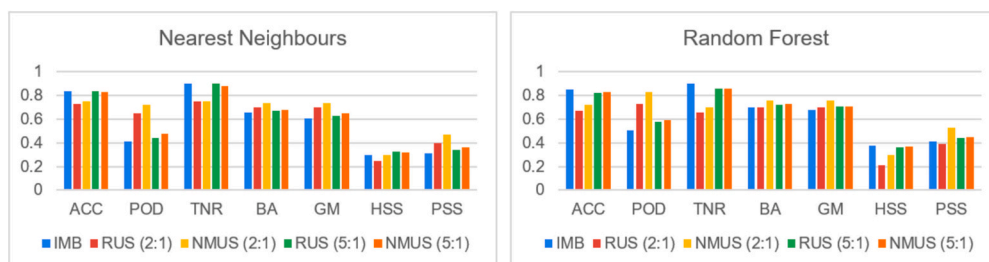


Fig. 5. Performance of Nearest Neighbours (NN) and Random Forest (RF) classifiers on test datasets of CT region for imbalanced datasets (IMB), Random Under-sampling (RUS), and Near-miss Undersampling (NMUS) with imbalance ratios 5:1 and 2:1.

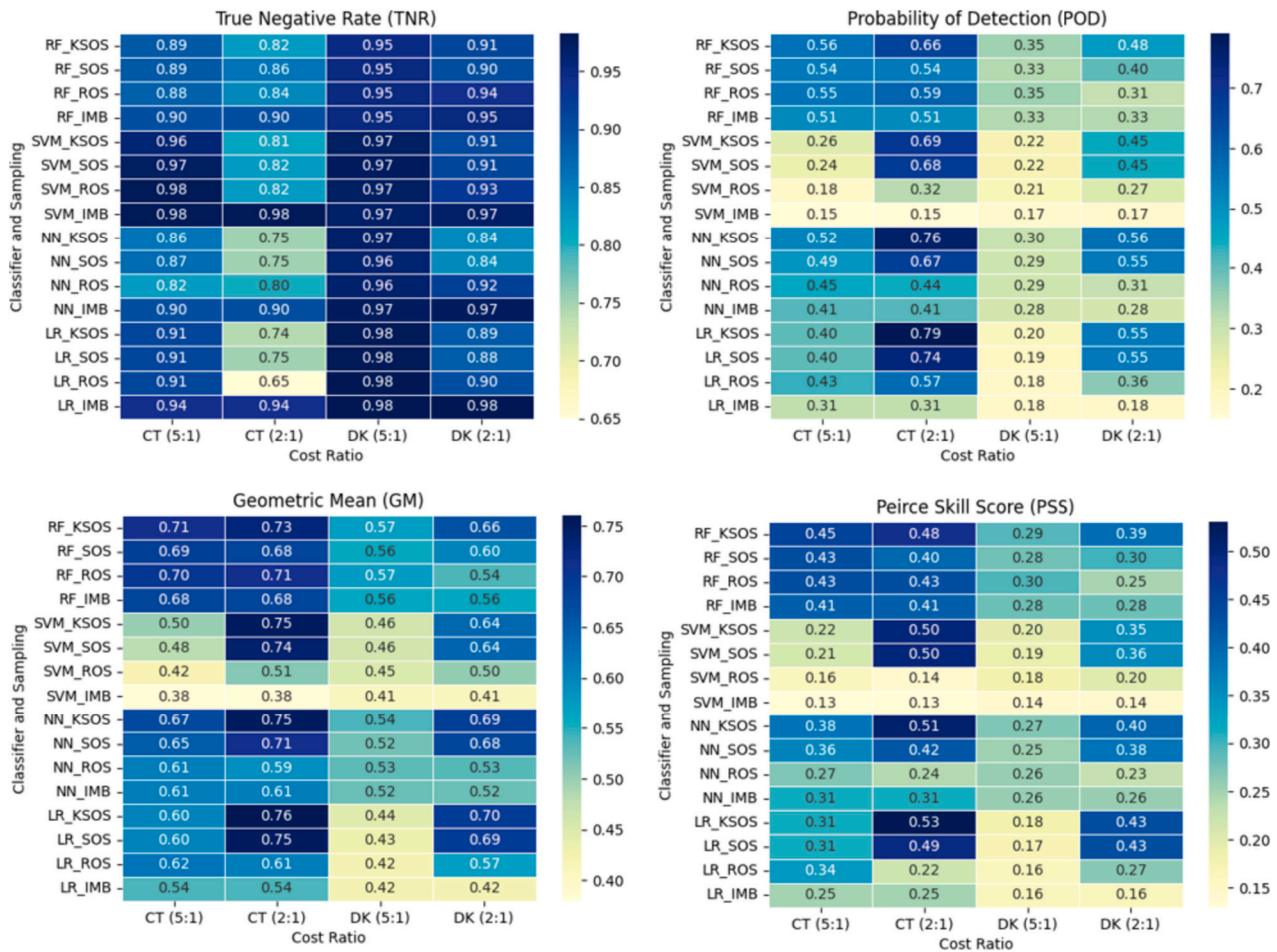


Fig. 6. Performance of classifiers in terms of TNR, POD, GM and PSS on the test dataset for imbalanced datasets (IMB), datasets resampled using random, SMOTE and KMeansSMOTE oversampling with imbalance ratios 5:1 and 2:1 on CT and DK regions.

worsening (decrease of around 20 %), thus resulting in poorer performance for non-avalanche days due to the fact that for the 2:1 ratio the sample size of the dominant non-avalanche class is decreased. Further, the already better performing Random Forest classifier also registered improvements in all metrics e.g. in CT region for training data resampled to 2:1 using NearMiss Undersampling it achieved Peirce Skill Score of 0.53, Geometric Mean of 0.76 and Heidke Skill Score of 0.3.

Comparing the performance of data resampled to IR 5:1 and 2:1 using NearMiss, the Peirce Skill Score values were 0.32 vs 0.50, 0.36 vs 0.47, 0.21 vs 0.47 and 0.45 vs 0.53 for Logistic Regression, Nearest Neighbours, Support Vector Machine, and Random Forest classifiers, respectively; similar comparable performance was found for Balanced Accuracy viz. 0.66 vs 0.75, 0.68 vs 0.74, 0.61 vs 0.74 and 0.73 vs 0.76 for the same classifiers for NearMiss Undersampling. Likewise, the performance was better for the ratio 2:1 for Random Undersampling too for most of the metrics.

In DK region, for IR 5:1 the Peirce Skill Score values for Logistic Regression and Support Vector Machine classifiers improved from 0.16 to 0.24 and 0.14 to 0.22 in case of NearMiss Undersampling; for Random Undersampling there was not much improvement in Probability of Detection, Balanced Accuracy and Geometric Mean values. In case of IR 2:1, for NearMiss Undersampling the Peirce Skill Score values increased to 0.52, 0.46, 0.35 and 0.45 for Logistic Regression, Nearest Neighbours, Support Vector Machine and Random Forest classifiers, respectively. Clearly, NearMiss Undersampling outperformed Random Undersampling for all classifiers across most evaluation metrics for both regions because its focused approach eliminated the non-avalanche instances to increase the spaces between the two classes, leading to

better classification.

6.3. Performance on oversampling avalanche day instances

In CT region, for the ratio 2:1 (Table A.4 and Fig. 6), the Probability of Detection values improved to around 0.6 for almost all classifiers (in the case of SMOTE and KMeansSMOTE, in particular); further, the Balanced Accuracy and Geometric Mean values neared 0.65 in most cases; still, the True Negative Rate was more than 0.75, thus meaning that the performance improved for the avalanche classes while not decreasing much for the non-avalanche days. A similar pattern was found in the case of oversampling to the 5:1 ratio, though the increase in Geometric Mean, Probability of Detection, and Peirce Skill Score was much lesser (Table A.3). For data resampled to IR 5:1 and 2:1 using KMeansSMOTE, the Peirce Skill Score values were 0.31 vs 0.53, 0.38 vs 0.51, 0.22 vs 0.50 and 0.45 vs 0.48 for Logistic Regression, Nearest Neighbours, Support Vector Machine, and Random Forest classifiers, respectively; similar comparable performance was found for Balanced Accuracy viz. 0.65 vs 0.76, 0.69 vs 0.75, 0.61 vs 0.75 and 0.73 vs 0.74 for the same classifiers using KMeansSMOTE Oversampling for the two ratios. The more balanced ratio of 2:1 was closer to equal class representation and made the models better at identifying the minority avalanche class instances.

For the DK region, there were more significant improvements in values of Probability of Detection, Geometric Mean, Peirce Skill Score and other metrics as compared to classifiers trained on imbalanced, e.g. Probability of Detection increased to more than 0.45 for all the classifiers in case of IR 2:1, the Peirce Skill Score values also improved from

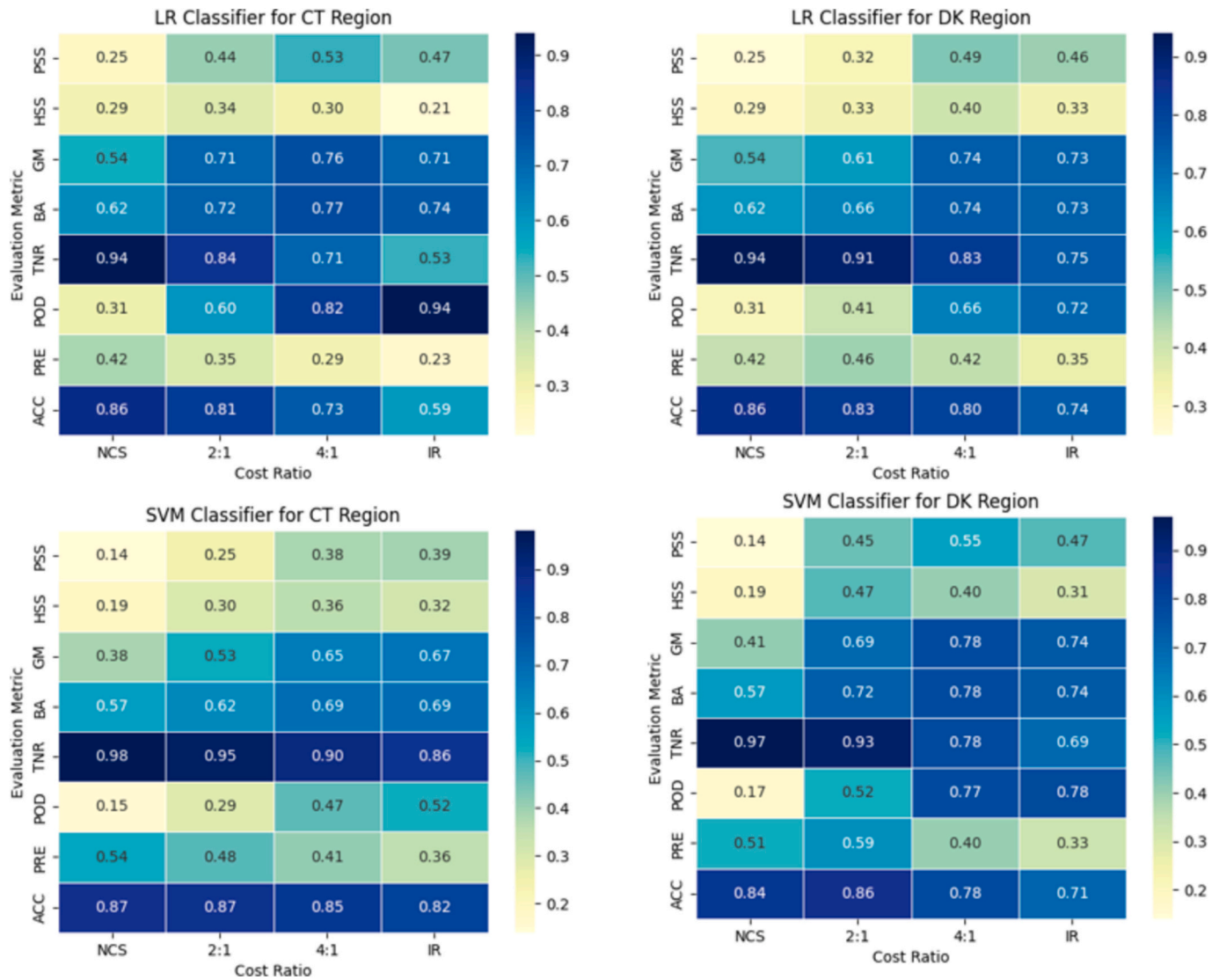


Fig. 7. Relative performance of cost-sensitive LR classifier and SVM classifier on cost ratios 2:1, 4:1 and the Inverse class ratio cost (IR) when compared to their Non Cost Sensitive (NCS) equivalents for CT and DK region.

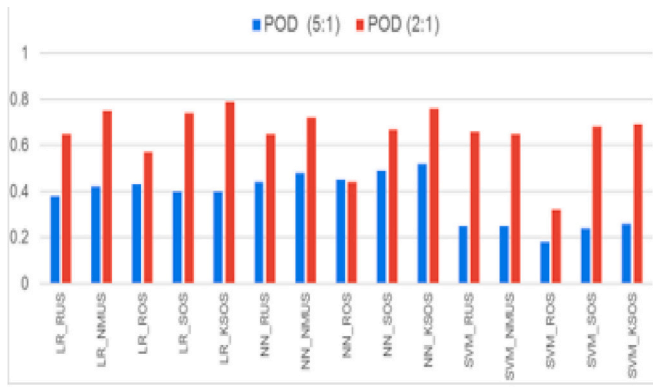
0.16 to 0.43, 0.26 to 0.40, 0.28 to 0.39 for Logistic Regression, Nearest Neighbours and Random Forest classifiers for data resampled to 2:1 ratio using KMeansSMOTE oversampling (Fig. 6).

In addition to a high Probability of Detection, a high True Negative Rate is also desirable because it signifies the instances belonging to non-avalanche days that were incorrectly classified as avalanche days by the classifiers. While all the classifiers trained on imbalanced data had a good True Negative Rate > 0.9 for both regions, the oversampling (introduction of additional avalanche days) did not decrease it much; when more oversampling was performed (IR 2:1), the average True Negative Rate was still > 0.75 , while for data oversampled to an imbalance ratio of 5:1 or 4:1, there was not much decrease in True Negative Rate. Further, the average Probability of Detection was higher for SMOTE and KMeansSMOTE oversampling than the Random Oversampling approach (Fig. 6); it is because the former approaches led to the generation of more 'realistic' additional avalanche days. The poor performance of Random Oversampling is due to the failure of classifiers trained on such 'copied' data to generalize well on the 'unseen' test data since they try to remember the oversampled data. So it may be concluded that oversampling leads to improvements in Probability of Detection, Geometric Mean, Peirce Skill Score, and Balanced Accuracy (improved performance on the minority avalanche class) without

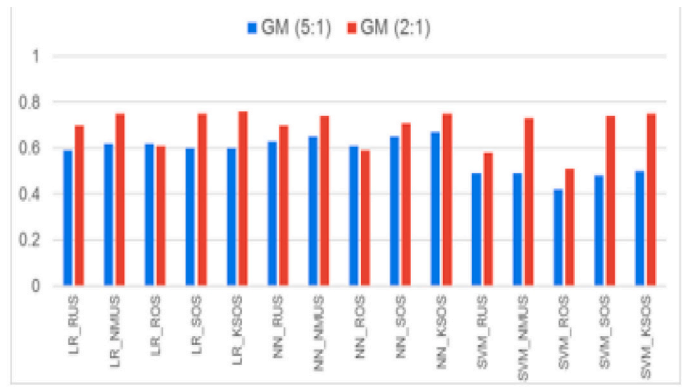
worsening the True Negative Rate values. The KMeansSMOTE was the clear winner (Fig. 6) among the three oversampling techniques for because of its ability to generate avalanche day instances in regions that are sparsely populated in the feature space for almost all classifiers for both regions in terms of different evaluation metrics.

6.4. Comparison of undersampling and oversampling approaches

It was found that generally oversampling resulted in better values of different evaluation metrics than undersampling for the same ratios since undersampling results in the loss of non-avalanche day instances. For IR of 2:1 and 5:1, the Probability of Detection values for all classifiers were better in cases of SMOTE and KMeansSMOTE oversampling than Random and NearMiss undersampling (Fig. 8a), since oversampling results in more avalanche instances during the training phase. However, the metrics that evaluate performance on both classes, like Geometric Mean (Fig. 8b), and Balanced Accuracy had comparable values for oversampling and undersampling. Clearly, KMeansSMOTE oversampling was the best performer for all the metrics across all classifiers, while Random Undersampling was the worst one since it randomly removed the non-avalanche instances.



(a) Probability of Detection (POD)



(b) Geometric Mean (GM)

Fig. 8. Comparison of classifiers' performance trained on datasets resampled using RUS, NMUS, ROS, SOS and KSOS for IR of 5:1 and 2:1 in terms of (a) POD and (b) GM on test datasets for CT region.

6.5. Performance of cost-sensitive classifiers

The cost-sensitive version of Random Forest Classifier exhibited similar or marginally improved values for the various evaluation criteria (Accuracy, Peirce Skill Score, Probability of Detection, etc.) with different cost ratios. This observation is in coherence with [Sielenou et al., 2021](#) who also reported that the class-balanced Random Forest did not perform well for the minority avalanche days; data resampling is more helpful in case of Random Forest classifiers. As shown in [Fig. 7](#) and [Table A.5](#), the cost-sensitive Logistic Regression classifier for CT region trained on inverse class distribution ($LR_{CS_{IR}}$) displayed a marked improvement in Probability of Detection (0.94 vs. 0.31), Balanced Accuracy (0.74 vs. 0.62), Geometric Mean (0.71 vs. 0.54), and Peirce Skill Score (0.47 vs. 0.25) when compared to its non-cost-sensitive equivalent (LR_{NCS}). Quite understandably, there was some deterioration in values for Accuracy and True Negative Rate due to more emphasis on avalanche class. For the other cost ratios of 4:1 (LR_{CS_4}) and 2:1 (LR_{CS_2}) in CT region, similar improvements were obtained at a marginal cost of lesser True Negative Rate values. The cost-sensitive Support Vector Machine classifier (SVM_{CS}) also exhibited similar results for these cost ratios ([Fig. 7](#)); however, the improvements in values were less significant than those of the Logistic Regression classifier (LR_{CS}) since the soft-margin hyperplane separating the two classes is less sensitive to these costs. Also, the improved performance of the SVM_{CS} for the avalanche class in terms of Probability of Detection, Balanced Accuracy, Peirce Skill Score, and Geometric Mean did not result in much worsening of performance on the non-avalanche class (True Negative Rate). Similar improvements in values of Peirce Skill Score, Geometric Mean, Balanced Accuracy and Probability of Detection were achieved for Logistic Regression and Support Vector Machine classifiers in the DK region; Peirce Skill Score improved to 0.46 and 0.39 for Logistic Regression and Support Vector Machine classifiers, respectively.

It may be concluded the more weightage one assigns to the false negatives than the false positives, the better the Probability of Detection since the classifiers now aim for better performance on the avalanche class, but the performance on the non-avalanche class may decline. For both these classifiers, optimal values for evaluation metrics other than Probability of Detection are obtained for the cost ratio of 4:1 rather than for IR:1 ([Fig. 7](#)). While giving too low misclassification costs to the avalanche class (e.g., 2:1) may prevent classification boundaries from adjusting properly, too high costs for the minority class (e.g., IR) may lead to poor generalization on the non-avalanche class. In situations of class imbalance, improperly set costs can inadvertently shift a bias from the majority non-avalanche class to a bias favouring the minority avalanche class. However, the goal should be to achieve balanced

performance for both classes; hence, cost ratios like 4:1 offer a decent trade-off between the performance on the two classes rather than higher cost ratios like IR.

6.6. Performance of classifier for CT and DK regions

The better performance observed in the CT region for both resampling and cost-sensitive approaches may be partially attributed to a lower class imbalance compared to the DK region. However, it is important to recognize that several other factors could influence these results. Differences in the climatic conditions, dataset characteristics, and the nature of input variables could significantly affect model performance. For instance, the DK region, characterized by a continental climate, may present unique challenges for avalanche prediction. This is because continental climates are typically associated with more variable weather patterns, which might introduce higher levels of uncertainty in the weather input data utilized in this study. Additionally, regional differences in terrain, snowpack characteristics, and the resolution or quality of the datasets could further contribute to the observed variations in performance.

6.7. Validity and generalizability of the results

The validity of the results is extremely important in such empirical studies. All experiments were conducted using the scikit-learn Python library ([Pedregosa et al., 2011](#)). The programs were thoroughly tested to remove all bugs that may adversely affect the reproducibility of results. The use of five-fold cross-validation prevents overfitting on training data and enhances the generalizability. Since multiple classifiers trained on data resampled using different techniques showed improvements in most of the parameters for both regions (with varying class imbalance), it builds the faith of the authors in the reliability of the experimental setup.

The approaches can also be extended to prediction of other rare events, however, results may vary when applied to different datasets with varying features; however, we expect improvements in classifiers' performance given the inherent mechanisms of these approaches.

6.8. Comparison with previous studies

Earlier attempts at class imbalance in avalanche forecasting did include some approaches (mostly random oversampling and a few on misclassification costs), but our attempt is quite extensive and varied in terms of approaches and classification paradigms studied. Our results for improvements in evaluation metrics compare well with the work of

Möhle et al., 2014 who trained Random Forest on an equal number of avalanche and non-avalanche boot-strapped instances, and weighted Random Forest with cost ratio 5:1; but they did not attempt different imbalance ratios. Möhle et al. (2014) reported Probability of Detection of 0.5, True Negative Rate of 0.95 for weighted Random Forest while we achieved Probability of Detection and True Negative Rate of 0.73 and 0.73, respectively for Random Forest Classifier trained with data resampled to 2:1 using NearMiss Undersampling for DK region; these values were 0.66 and 0.82 for Random Forest Classifier trained with data resampled to 2:1 using KMeansSMOTE Oversampling for CT region. Like our study, Sielenou et al., 2021 used multiple classifiers like linear discriminant analysis, classification trees, Support Vector Machine and Random Forest with oversampling to discriminate between three classes of avalanche activities and reported improvement in misclassification rates; however, they employed only random oversampling to achieve IR of 1:1. Since we considered a rather lenient criterion to consider all avalanches into one class, our results are not directly comparable to those of Sielenou et al., 2021 which considers different types of avalanches as distinct classes. The Heidke Skill Score values of 0.32 and 0.30 for Nearest Neighbours classifier trained on dataset resampled to IR 2:1 using KMeansSMOTE Oversampling and NearMiss Undersampling, respectively of our results are comparable to 0.31 reported by Singh et al., 2015 in CT region and Heidke Skill Score values of around 0.34 for Nearest Neighbours classifier trained on dataset resampled to IR 2:1 using KMeansSMOTE Oversampling and to IR 2:1 using NearMiss Undersampling for DK region compare favourably with 0.27 reported by Singh et al., 2015 in the same region using Nearest Neighbours classifier. The Heidke Skill Score values for our other classifiers are better than those reported by Singh et al., 2015.

6.9. Limitations and operational forecasting

There are some limitations of the present study. The non-avalanche days used in the study may have witnessed avalanche activities that were unreported, this can have serious implications for performance of the classifiers discussed. Secondly, we defined an avalanche day based on occurrence of any avalanches, regardless of their type, size or other characteristics; since dividing them into subtypes would have exacerbated the class imbalance. However, Mayer et al., 2023 suggest that distinct classifiers may be necessary for different avalanche types due to variations in their dynamics, environmental conditions, etc. In our study, we focused on basic snow-meteorological variables that have been measured across the semi-automated and manual observatories over the Indian Himalayas to establish a baseline understanding of the impact of class imbalance. However, recent studies (Pérez-Guillén et al., 2022; Viallon-Galinier et al., 2023) suggest that advanced snow stability variables and snow cover modelling can significantly enhance predictive performance. We are sure that including these advanced snow stability variables in conjunction with class imbalance approaches would lead to much better results in terms of improvements in the evaluation metrics.

A limitation of the present work for operational use is that the outcomes elaborated pertain to the test datasets, that contained recorded parameter values. Consequently, the evaluation provided pertains to retrospective data analysis. In forthcoming endeavours, this methodology could be integrated with appropriate weather prediction models, enabling operational avalanche forecasting within the Himalayan region by leveraging forecasted parameter values. In the context of climate change, avalanche forecasting models face challenges such as concept drift (when the relationship between input features and output predictions changes) and data drift (when the statistical properties of input data evolve over time). To ensure the robustness and reliability of forecasting models under these conditions, we propose to retrain the models periodically using the most recent data and also track the models' performance metrics over time.

7. Conclusion

In this paper, we presented a comprehensive investigation into the impact of class imbalance on avalanche forecasting. Our analysis encompassed a detailed exploration of various strategies employed to address this imbalance, a technical assessment of multiple evaluation metrics, and extensive empirical investigations. Our experimental design included a wide spectrum of scenarios, encompassing varying degrees of class imbalance. This allowed us to assess the efficacy of diverse classifiers across these imbalance ratios. Moreover, we delved into the performance of cost-sensitive adaptations of these classifiers, adding an extra layer of insight into their practical utility.

From the results it is evident that data resampling does indeed improve the performance of all classifiers, as is apparent from different evaluation criteria like Probability of Detection, Balanced Accuracy, Geometric Mean, Heidke Skill Score, Peirce Skill Score, etc. The KMeansSMOTE oversampling demonstrated better values for most of the evaluation metrics when compared to other resampling techniques since it generates synthetic avalanche day instances that are more credible by virtue of being closer to existing instances.

Cost-sensitive techniques that give more importance to positive instances (avalanche days) than negative instances also reported significant improvements in almost all evaluation criteria. The different data resampling and cost-sensitive paradigms evaluated in the present study result in an improvement of Probability of Detection for most of the classifiers (the desirable effect of detecting more avalanche days) without a significant decrease in the True Negative Rate (the undesirable effect of predicting the non-avalanche days as avalanche days).

Comparing the approaches of oversampling, undersampling and cost-sensitive learning; in our results we found that different classifiers responded differently to various sampling techniques, reporting different degrees of improvement in various evaluation criteria. Generally, oversampling techniques are beneficial when dealing with severe class imbalance (e.g. IR 13.3:1 in DK) as there are very less instances of the avalanche class for the models to 'learn' the underlying characteristics. Cost-sensitive approaches are better when resampling is not feasible, particularly in cases where the dataset is too small or when maintaining the original distribution of the data is crucial. For the development of classification models, when the absolute number of avalanche class instances are very less, we should prefer KMeansSMOTE oversampling for all classifiers. Further, the over- and under-sampling can be performed together on the same dataset for better predictive performance. For multi-class classifiers (e.g. for different types of avalanches), data resampling is preferable since it is difficult to assign comparative costs to more than two classes and it is not feasible to calculate the inverse cost ratios.

8. Future work

Since the class imbalance problem has only recently been addressed in avalanche forecasting, there is immense potential in the area. The paper suggests different sampling techniques to mitigate the class imbalance, we plan to combine oversampling and undersampling approaches for a hybrid approach that may give better results. Another course of action is the use of ensemble approaches that combine multiple classifiers trained on balanced datasets for a comprehensive decision support system for forecasting avalanches in the Indian Himalayas. For imbalanced datasets, the standard classification threshold of 0.5 is suboptimal; in future investigations, it might be worthwhile to explore the technique of threshold tuning that involves determining the most suitable thresholds using metrics such as ROC (Receiver Operating Characteristic) or Precision and Recall curves. Study by Maloof, 2003 demonstrated that sampling the classes gives similar results as moving threshold boundary, still we can explore threshold tuning as a potential avenue for enhancing classification performance. It is also worthwhile to analyse avalanches that differ in terms of sizes and types as separate

classes and employ the resampling approaches to mitigate class imbalance within these distinct avalanche types to enhance the predictive power of our models.

Lastly, the paper has discussed only the issue of relative imbalance in the avalanche and non-avalanche classes due to their different frequencies in the datasets; other issues like the overlap between the instances of the two classes, sample size of the avalanche class (hence the problem of learning the underlying characteristics of the avalanche class), noisy instances, and small disjuncts (sub-concepts within the avalanche class) can also be addressed in future work.

CRedit authorship contribution statement

Manish Kala: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Shweta Jain:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Amreek Singh:** Writing – review & editing, Visualization, Methodology. **Narayanan Chatapuram Krishnan:** Writing – review & editing,

Writing – original draft, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgements

The authors acknowledge the DGRE personnel for their valuable contribution to the collection of snow and meteorological data across multiple observatories in the harsh climatic conditions of the Himalayas. The technical suggestions and critical comments shared by colleagues are also duly acknowledged.

Appendix A. Full result tables and optimized parameters

This appendix reports the full results for the performance of various classifiers on the original imbalanced datasets, datasets resampled using different undersampling and oversampling approaches, cost-sensitive classifiers and also finalized hyper-parameters for the classifiers. The convention used is Model_SamplingApproach e.g. LR_IMB denotes Logistic Regression Classifier's performance on imbalanced dataset, LR_RUS denotes Logistic Regression Classifier's performance on data resampled using Random Undersampling (RUS). Each table entry has the absolute value of the evaluation criteria for sampling technique considered. The performance for the imbalanced datasets is in bold for all classifiers.

Table A.1

Performance on test datasets of classifiers trained using imbalanced datasets (IMB), datasets with IR of 5:1 resampled through Random Undersampling (RUS) and Near-miss Undersampling (NMUS) for CT region.

Classifier and sampling	Evaluation Metric						
	ACC	POD	TNR	BA	GM	HSS	PSS
LR_IMB	0.86	0.31	0.94	0.62	0.54	0.29	0.25
LR_RUS	0.85	0.38	0.92	0.65	0.59	0.31	0.30
LR_NMUS	0.84	0.42	0.90	0.66	0.62	0.31	0.32
NN_IMB	0.84	0.41	0.90	0.66	0.61	0.30	0.31
NN_RUS	0.84	0.44	0.90	0.67	0.63	0.33	0.34
NN_NMUS	0.83	0.48	0.88	0.68	0.65	0.32	0.36
SVM_IMB	0.87	0.15	0.98	0.57	0.38	0.19	0.13
SVM_RUS	0.87	0.25	0.97	0.61	0.49	0.29	0.22
SVM_NMUS	0.87	0.25	0.96	0.61	0.49	0.27	0.21
RF_IMB	0.85	0.51	0.90	0.70	0.68	0.38	0.41
RF_RUS	0.82	0.58	0.86	0.72	0.71	0.36	0.44
RF_NMUS	0.83	0.59	0.86	0.73	0.71	0.37	0.45

Table A.2

Performance on test datasets of classifiers trained using imbalanced datasets (IMB), datasets with IR of 2:1 resampled through Random Undersampling (RUS) and Near-miss Undersampling (NMUS) for CT region.

Classifier and sampling	Evaluation Metric						
	ACC	POD	TNR	BA	GM	HSS	PSS
LR_IMB	0.86	0.31	0.94	0.62	0.54	0.29	0.25
LR_RUS	0.73	0.65	0.74	0.70	0.7	0.25	0.39
LR_NMUS	0.75	0.75	0.75	0.75	0.75	0.31	0.50
NN_IMB	0.84	0.41	0.9	0.66	0.61	0.30	0.31
NN_RUS	0.73	0.65	0.75	0.70	0.70	0.25	0.40
NN_NMUS	0.75	0.72	0.75	0.74	0.74	0.30	0.47
SVM_IMB	0.87	0.15	0.98	0.57	0.38	0.19	0.13
SVM_RUS	0.53	0.66	0.51	0.59	0.58	0.08	0.17
SVM_NMUS	0.80	0.65	0.82	0.74	0.73	0.35	0.47
RF_IMB	0.85	0.51	0.90	0.70	0.68	0.38	0.41
RF_RUS	0.67	0.73	0.66	0.70	0.70	0.21	0.39
RF_NMUS	0.72	0.83	0.70	0.76	0.76	0.30	0.53

Table A.3

Performance on test datasets of classifiers trained using imbalanced datasets (IMB), datasets with IR of 5:1 resampled through Random Oversampling (ROS), SMOTE Oversampling (SOS) and KMeansSMOTE Oversampling (KSOS) approaches for CT region.

Classifier and sampling	Evaluation Metric						
	ACC	POD	TNR	BA	GM	HSS	PSS
LR_IMB	0.86	0.31	0.94	0.62	0.54	0.29	0.25
LR_ROS	0.85	0.43	0.91	0.67	0.62	0.33	0.34
LR_SOS	0.84	0.40	0.91	0.65	0.60	0.31	0.31
LR_KSOS	0.84	0.40	0.91	0.65	0.60	0.31	0.31
NN_IMB	0.84	0.41	0.9	0.66	0.61	0.30	0.31
NN_ROS	0.77	0.45	0.82	0.64	0.61	0.22	0.27
NN_SOS	0.82	0.49	0.87	0.68	0.65	0.31	0.36
NN_KSOS	0.81	0.52	0.86	0.69	0.67	0.32	0.38
SVM_IMB	0.87	0.15	0.98	0.57	0.38	0.19	0.13
SVM_ROS	0.87	0.18	0.98	0.58	0.42	0.23	0.16
SVM_SOS	0.87	0.24	0.97	0.60	0.48	0.28	0.21
SVM_KSOS	0.87	0.26	0.96	0.61	0.50	0.28	0.22
RF_IMB	0.85	0.51	0.90	0.70	0.68	0.38	0.41
RF_ROS	0.84	0.55	0.88	0.71	0.70	0.38	0.43
RF_SOS	0.85	0.54	0.89	0.72	0.69	0.39	0.43
RF_KSOS	0.85	0.56	0.89	0.73	0.71	0.40	0.45

Table A.4

Performance on test datasets of classifiers trained using imbalanced datasets (IMB), datasets with IR of 2:1 resampled through Random Oversampling (ROS), SMOTE Oversampling (SOS) and KMeansSMOTE Oversampling (KSOS) approaches for CT region.

Classifier and sampling	Evaluation Metric						
	ACC	POD	TNR	BA	GM	HSS	PSS
LR_IMB	0.86	0.31	0.94	0.62	0.54	0.29	0.25
LR_ROS	0.64	0.57	0.65	0.61	0.61	0.12	0.22
LR_SOS	0.75	0.74	0.75	0.75	0.75	0.31	0.49
LR_KSOS	0.75	0.79	0.74	0.76	0.76	0.32	0.53
NN_IMB	0.84	0.41	0.90	0.66	0.61	0.30	0.31
NN_ROS	0.75	0.44	0.80	0.62	0.59	0.18	0.24
NN_SOS	0.74	0.67	0.75	0.71	0.71	0.27	0.42
NN_KSOS	0.75	0.76	0.75	0.75	0.75	0.32	0.51
SVM_IMB	0.87	0.15	0.98	0.57	0.38	0.19	0.13
SVM_ROS	0.76	0.32	0.82	0.57	0.51	0.12	0.14
SVM_SOS	0.80	0.68	0.82	0.75	0.74	0.36	0.50
SVM_KSOS	0.80	0.69	0.81	0.75	0.75	0.36	0.50
RF_IMB	0.85	0.51	0.90	0.70	0.68	0.38	0.41
RF_ROS	0.81	0.59	0.84	0.72	0.71	0.34	0.43
RF_SOS	0.82	0.54	0.86	0.70	0.68	0.33	0.40
RF_KSOS	0.80	0.66	0.82	0.74	0.73	0.35	0.48

Table A.5

Performance of cost-sensitive LR and SVM classifiers on test dataset of CT region for different cost ratios. LR_NCS, LR_CS₂, LR_CS₄ and LR_CS_{IR} represent non-cost-sensitive LR classifier; LR classifier trained with cost ratios of 2:1, 4:1 and inverse class distribution (IR) respectively. The values for non-cost-sensitive classifiers are in bold.

Classifier and Cost Ratio	Evaluation Metric						
	ACC	POD	TNR	BA	GM	HSS	PSS
LR_NCS	0.86	0.31	0.94	0.62	0.54	0.29	0.25
LR_CS ₂	0.81	0.6	0.84	0.72	0.71	0.34	0.44
LR_CS ₄	0.73	0.82	0.71	0.77	0.76	0.3	0.53
LR_CS _{IR}	0.59	0.94	0.53	0.74	0.71	0.21	0.47
SVM_NCS	0.87	0.15	0.98	0.57	0.38	0.19	0.14
SVM_CS ₂	0.87	0.29	0.95	0.62	0.53	0.30	0.25
SVM_CS ₄	0.85	0.47	0.90	0.69	0.65	0.36	0.38
SVM_CS _{IR}	0.82	0.52	0.86	0.69	0.67	0.32	0.39

Table A.6
Performance of cost-sensitive LR and SVM classifiers on test dataset of DK region for different cost ratios. The values for non-cost-sensitive classifiers are in bold.

Classifier and Cost Ratio	Evaluation Metric						
	ACC	POD	TNR	BA	GM	HSS	PSS
LR_NCS	0.86	0.31	0.94	0.62	0.54	0.29	0.25
LR_CS ₂	0.83	0.41	0.91	0.66	0.61	0.33	0.32
LR_CS ₄	0.80	0.66	0.83	0.74	0.74	0.40	0.49
LR_CS _{IR}	0.74	0.72	0.75	0.73	0.73	0.33	0.46
SVM_NCS	0.87	0.15	0.98	0.57	0.38	0.19	0.14
SVM_CS ₂	0.87	0.29	0.95	0.62	0.53	0.30	0.25
SVM_CS ₄	0.85	0.47	0.90	0.69	0.65	0.36	0.38
SVM_CS _{IR}	0.82	0.52	0.86	0.69	0.67	0.32	0.39

Table A.7
Final hyper-parameters selected for the different optimized models for CT region.

Classification Model	Hyper Parameter	Keyword in sklearn	Finalized Value
Logistic Regression (LR)	Regularization parameter	penalty	L2
	Tolerance for stopping criteria	tol	0.0001
	Maximum number of iterations permitted for convergence	max_itr	100
Nearest Neighbours (NN)	Number of neighbours to consider for each query point	n_neighbors	10
	Weights for different parameters	weights	uniform
	Distance metric	metric and p	p = 2 i.e. Euclidean distance
Support Vector Machine (SVM)	L2 regularization parameter	C	0.5
	Kernel type to be used	kernel	rbf i.e. Radial Basis Function
Random Forest (RF)	Number of trees in forest	n_estimators	500
	Maximum number of features considered for splitting a node	max_features	log ₂ i.e. log ₂ (no. of features)
	Criterion used to measure quality of split	criterion	gini
	Maximum number of levels in each decision tree	max_depth	50
	Method for sampling data points (with or without replacement)	bootstrap	True i.e. sample with replacement

References

Ali, A., Shamsuddin, S.M., Ralescu, A.L., 2013. Classification with class imbalance problem. *Int. J. Adv. Soft Comput. Appl.* 5 (3), 176–204. <https://api.semanticscholar.org/CorpusID:26644563>.

Batuwita, R., Palade, V., 2013. Class imbalance learning methods for support vector machines. In: He, H., Ma, Y. (Eds.), *Imbalanced Learning*, pp. 83–99. <https://doi.org/10.1002/9781118646106.ch5>.

Bishop, C. M., 2016. *Pattern Recognition and Machine Learning*, First ed. Springer, New York. ISBN 978–0–387–31073–2.

Breiman, L., 2001. Random forests. *Mach. Learn.* 45, 5–32. <https://doi.org/10.1023/A:1010933404324>.

Brodersen, K.H., Ong, C.S., Stephan, K.E., Buhmann, J.M., 2010. The balanced accuracy and its posterior distribution. In: *Proceedings of the 20th International Conference on Pattern Recognition*, pp. 3121–3124. <https://doi.org/10.1109/ICPR.2010.764>.

Buser, O., 1983. Avalanche forecasting with the method of nearest neighbours: an interactive approach. *Cold Reg. Sci. Technol.* 8 (2), 155–163. [https://doi.org/10.1016/0165-232X\(83\)90006-X](https://doi.org/10.1016/0165-232X(83)90006-X).

Chawla, N.V., Bowyer, K.W., Hall, L.O., Kegelmeyer, W.P., 2002. SMOTE: synthetic minority over-sampling technique. *J. Artif. Intell.* 16, 321–357. <https://doi.org/10.1613/jair.953>.

Cortes, C., Vapnik, V., 1995. Support-vector networks. *Mach. Learn.* 20, 273–297. <https://doi.org/10.1007/BF00994018>.

Douzas, G., Bacao, F., Last, F., 2018. Improving imbalanced learning through a heuristic oversampling method based on k-means and SMOTE. *Inf. Sci.* 465, 1–20. <https://doi.org/10.1016/j.ins.2018.06.056>.

Ebert, P.A., Milne, P., 2022. Methodological and conceptual challenges in rare and severe event forecast verification. *Nat. Hazards Earth Syst. Sci.* 22, 539–557. <https://doi.org/10.5194/nhess-22-539-2022>.

Eckerstorfer, M., Malnes, E., Vickers, H., Müller, K., Engeset, R., Humstad, T., 2018. Operational avalanche activity monitoring using radar satellites: from norway to worldwide assistance in avalanche forecasting. In: *Proceedings of International Snow Science Workshop, Innsbruck, Austria*.

Ekmeckioglu, O., Koc, K., Özger, M., Işık, Z., 2022. Exploring the additional value of class imbalance distributions on interpretable flash flood susceptibility prediction in the Black Warrior River basin, Alabama, United States. *J. Hydrol.* 610 (127877). <https://doi.org/10.1016/j.jhydrol.2022.127877>.

Gassner, M., Etter, H., Birkeland, K., 2001. NXD2000 - an improved avalanche forecasting program based on the nearest neighbor method. In: *Proceedings of the International Snow Science Workshop*, pp. 52–59.

Haegeli, P., McClung, D.M., 2007. Expanding the snow-climate classification with avalanche-relevant information: initial description of avalanche winter regimes forsouthwestern Canada. *J. Glaciol.* 53 (181), 266–276. <https://doi.org/10.3189/172756507782202801>.

Haibo, H., Ma, Y., 2013. *Imbalanced Learning: Foundations, Algorithms, and Applications*, First edn. Wiley-IEEE Press. <https://doi.org/10.1002/9781118646106>.

Haixiang, G., Yijing, L., Shang, J., Mingyun, G., Yuanyue, H., Bing, G., 2017. Learning from class imbalanced data: Review of methods and applications. *Expert Syst. Appl.* 73, 220–239. <https://doi.org/10.1016/j.eswa.2016.12.035>.

Harrington, P., 2012. *Machine Learning in Action*, First ed. Manning Publications, New York. ISBN 9781617290183.

He, H., Garcia, E.A., 2009. Learning from imbalanced data. *IEEE Trans. Knowl. Data Eng.* 21 (9), 1263–1284. <https://doi.org/10.1109/TKDE.2008.239>.

Hendrikx, J., Murphy, M., Onslow, T., 2014. Classification trees as a tool for operational avalanche forecasting on the Seward highway, Alaska. *Cold Reg. Sci. Technol.* 97, 113–120. <https://doi.org/10.1016/j.coldregions.2013.08.009>.

Horton, S., Towell, M., Pascal, H., 2020. Examining the operational use of avalanche problems with decision trees and model-generated weather and snowpack variables. *Nat. Hazards Earth Syst. Sci.* 20 (12), 3551–3576. <https://doi.org/10.5194/nhess-20-3551-2020>.

Hosseinzadeh, M., Eftekhari, M., 2015. Improving rotation forest performance for imbalanced data classification through fuzzy clustering. In: *Proceedings of the International Symposium on Artificial Intelligence and Signal Processing (AISP)*. Mashhad, Iran, pp. 35–40. <https://doi.org/10.1109/AISP.2015.7123535>.

Johnson, J.M., Khoshgoftaar, T.M., 2019. Survey on deep learning with class imbalance. *J. Big Data* 6, 27. <https://doi.org/10.1186/s40537-019-0192-5>.

Joshi, J.C., Srivastava, S., 2014. A Hidden Markov Model for avalanche forecasting on Chowkibal-Tangdhar road axis in Indian Himalayas. *J. Earth Syst. Sci.* 123 (8), 1771–1779. <https://doi.org/10.1007/s12040-014-0510-4>.

Krawczyk, B., Galar, M., Jelen, Ł., Herrera, F., 2016. Evolutionary undersampling boosting for imbalanced classification of breast cancer malignancy. *Appl. Soft Comput.* 38, 714–726. <https://doi.org/10.1016/j.asoc.2015.08.060>.

LaChapelle, E.R., 1980. The fundamental processes in conventional avalanche forecasting. *J. Glaciol.* 26 (94), 75–84. <https://doi.org/10.3189/S0022143000010601>.

Liu, X.Y., Wu, J., Zhou, Z.H., 2008. Exploratory undersampling for class-imbalance learning. *IEEE Trans. Syst. Man Cybern. B Cybern.* 39 (2), 539–550. <https://doi.org/10.1109/TSMCB.2008.2007853>.

Maldonado, S., López, J., Vairetti, C., 2019. An alternative SMOTE oversampling strategy for high-dimensional datasets. *Appl. Soft Comput.* 76, 380–389. <https://doi.org/10.1016/j.asoc.2018.12.024>.

Malooof, M.A., 2003. Learning when data sets are imbalanced and when costs are unequal and unknown. In: *ICML-2003 Workshop on Learning From Imbalanced Data Sets II*.

- Mani, I., Zhang, I., 2003. kNN approach to unbalanced data distributions: A case study involving information extraction. In: Proceedings of the ICML 2003 Workshop on Learning From Imbalanced Datasets. <https://cit.rii.ac.jp/crid/157423187400444032>.
- Marienthal, A., Hendrikx, J., Birkeland, K., Irvine, K.M., 2015. Meteorological variables to aid forecasting deep slab avalanches on persistent weak layers. *Cold Reg. Sci. Technol.* 120, 227–236. <https://doi.org/10.1016/j.coldregions.2015.08.007>.
- Mayer, S., Herwijnen, A.V., Olivieri, G., Schweizer, J., 2020. Evaluating the performance of an operational infrasound avalanche detection system at three locations in the Swiss Alps during two winter seasons. *Cold Reg. Sci. Technol.* 173, 102962. <https://doi.org/10.1016/j.coldregions.2019.102962>.
- Mayer, S., Techel, F., Schweizer, J., van Herwijnen, A., 2023. Prediction of natural dry-snow avalanche activity using physics-based snowpack simulations. *Nat. Hazards Earth Syst. Sci.* 23 (11), 3445–3465. <https://doi.org/10.5194/nhess-23-3445-2023>.
- McClung, D.M., 2000. Predictions in avalanche forecasting. *Ann. Glaciol.* 31, 377–381. <https://doi.org/10.3189/172756400781820507>.
- McClung, D.M., 2002. The elements of applied avalanche forecasting, Part II: the physical issues and the rules of applied avalanche forecasting. *Nat. Hazards* 26 (2), 131–146. <https://doi.org/10.1023/A:1015665432221>.
- McCollister, C., Birkeland, K.W., Hansen, K., Aspinall, R., 2003. Exploring multi-scale spatial patterns in historical avalanche data. Jackson Hole Ski Area Wyom., *Cold Regions Sci. Technol.* 37 (3), 299–313. [https://doi.org/10.1016/S0165-232X\(03\)00072-7](https://doi.org/10.1016/S0165-232X(03)00072-7).
- Moepya, S.O., Akhouri, S.S., Nelwamondo, F.V., 2014. Applying cost-sensitive classification for financial fraud detection under high class-imbalance. Proceedings of the IEEE International Conference on Data Mining Workshops, ICDM Workshops 2014, Shenzhen, pp. 183–192. doi: <https://doi.org/10.1109/ICDMW.2014.141>.
- Möhle, S., Bründl, M., Beierle, C., 2014. Modeling a system for decision support in snow avalanche warning using balanced random forest and weighted random forest. In: Agre, G., Hitzler, P., Krisnadhi, A.A., Kuznetsov, S.O. (Eds.), *Artificial Intelligence: Methodology, Systems, and Applications. AIMS 2014, Lecture Notes in Computer Science*, Vol. 8722. Springer, Cham. https://doi.org/10.1007/978-3-319-10554-3_8.
- Osorio, J.H., Meza, A.A., Santacoloma, G.D., Gutierrez, A.O., Dominguez, G.C., 2021. Relevant information undersampling to support imbalanced data classification. *Neurocomputing* 436, 136–146. <https://doi.org/10.1016/j.neucom.2021.01.033>.
- Pant, L.M., Ganju, A., 2006. Application of neural networks for avalanche forecasting. *Def. Sci. J.* 56 (4), 559–567. <https://doi.org/10.14429/dsj.56.1921>.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, E., 2011. Scikit-learn: machine learning in python. *J. Mach. Learn. Res.* 12, 2825–2830.
- Pérez-Guillén, C., Techel, F., Hendrick, M., Volpi, M., Herwijnen, A.V., Olevski, T., Obózinski, G., Cruz, F.P., Schweizer, J., 2022. Data-driven automated predictions of the avalanche danger level for dry-snow conditions in Switzerland. *Nat. Hazards Earth Syst. Sci.* 22 (6), 2031–2056. <https://doi.org/10.5194/nhess-22-2031-2022>.
- Pham, Q.B., Ekmekcioglu, O., Ali, S.K., Koc, K., Parvin, F., 2023. Examining the role of class imbalance handling strategies in predicting earthquake-induced landslide-prone regions. *Appl. Soft Comput.* 143, 110429. <https://doi.org/10.1016/j.asoc.2023.110429>.
- Pozdnoukhov, A., Purves, R.S., Kanevski, M., 2008. Applying machine learning methods to avalanche forecasting. *Ann. Glaciol.* 49, 107–113. <https://doi.org/10.3189/172756408787814870>.
- Pozdnoukhov, A., Matasci, G., Kanevski, M., Purves, R.S., 2011. Spatio-temporal avalanche forecasting with support vector machines. *Nat. Hazards Earth Syst. Sci.* 11, 367–382. <https://doi.org/10.5194/nhess-11-367-2011>.
- Purves, R.S., Morrison, K.W., Moss, G., Wright, D.S.B., 2003. Nearest neighbours for avalanche forecasting in Scotland - development, verification and optimisation of a model. *Cold Reg. Sci. Technol.* 37 (3), 343–355. [https://doi.org/10.1016/S0165-232X\(03\)00075-2](https://doi.org/10.1016/S0165-232X(03)00075-2).
- Rosenthal, W., Elder, K., Davis, E.R., 2002. Operational Decision Tree Avalanche Forecasting. International Snow Science Workshop, Penticton B.C., pp. 152–158.
- Schweizer, J., Mitterer, C., Techel, F., Stoffel, A., Reuter, B., 2020. On the relation between avalanche occurrence and avalanche danger level. *Cryosphere* 14, 737–750. <https://doi.org/10.5194/tc-14-737-2020>.
- Seiffert, C., Khoshgoftaar, T.M., Hulse, J.V., Napolitano, A., 2007. Mining data with rare events: a case study. In: Proceedings of the 19th IEEE International Conference on Tools With Artificial Intelligence (ICTAI 2007), pp. 132–139. <https://doi.org/10.1109/ICTAI.2007.71>.
- Sharma, S.S., Ganju, A., 2000. Complexities of avalanche forecasting in Western Himalayas - an overview. *Cold Reg. Sci. Technol.* 31, 95–102. [https://doi.org/10.1016/S0165-232X\(99\)00034-8](https://doi.org/10.1016/S0165-232X(99)00034-8).
- Sielenou, P.D., Viallon-Galinier, L., Hagenmuller, P., Naveau, P., Morin, S., Dumont, M., Verfaillie, D., Eckert, N., 2021. Combining random forests and class-balancing to discriminate between three classes of avalanche activity in the French Alps. *Cold Reg. Sci. Technol.* 187, 103276. <https://doi.org/10.1016/j.coldregions.2021.103276>.
- Singh, A., Ganju, A., 2004. A supplement to nearest-neighbour method for avalanche forecasting. *Cold Reg. Sci. Technol.* 39, 105–113. <https://doi.org/10.1016/j.coldregions.2004.03.005>.
- Singh, A., Srinivasan, K., Ganju, A., 2005. Avalanche forecast using numerical weather prediction in Indian Himalayas. *Cold Reg. Sci. Technol.* 43, 83–92. <https://doi.org/10.1016/j.coldregions.2005.05.009>.
- Singh, A., Damir, B., Deep, K., Ganju, A., 2015. Calibration of nearest neighbors model for avalanche forecasting. *Cold Reg. Sci. Technol.* 109, 33–42. <https://doi.org/10.1016/j.coldregions.2014.09.009>.
- Skamarock, W.C., Klemp, J.B., Dudhia, J., Gill, D.O., Liu, Z., Berner, J., Wang, W., Powers, J.G., Duda, M.G., Barker, D.M., Huang, X.-Y., 2021. A Description of the Advanced Research WRF Model Version 4.3 (No. NCAR/TN-556+STR). <https://doi.org/10.5065/1dth-6p97>.
- Snieder, E., Abogadil, K., Khan, U.T., 2021. Resampling and ensemble techniques for improving ANN-based high-flow forecast accuracy. *Hydrol. Earth Syst. Sci.* 25, 2543–2566. <https://doi.org/10.5194/hess-25-2543-2021>.
- Trafalis, T.B., Adrianto, I., Richman, M.B., Lakshminarayanan, S., 2014. Machine-learning classifiers for imbalanced tornado data. *Comput. Manag. Sci.* 11 (4), 403–418. <https://doi.org/10.1007/s10287-013-0174-6>.
- van Herwijnen, A., Schweizer, J., 2011. Monitoring avalanche activity using a seismic sensor. *Cold Reg. Sci. Technol.* 69, 165–176. <https://doi.org/10.1016/j.coldregions.2011.06.008>.
- Veropoulos, K., Campbell, C., Cristianini, N., 1999. Controlling the sensitivity of support vector machines. In: Proceedings of the International Joint Conference on Artificial Intelligence, Stockholm, Sweden, pp. 55–60.
- Viallon-Galinier, L., Hagenmuller, P., Eckert, N., 2023. Combining modelled snowpack stability with machine learning to predict avalanche activity. *Cryosphere* 17 (6), 2245–2260. <https://doi.org/10.5194/tc-17-2245-2023>.
- Vuttipittayamongkol, P., Elyan, E., Petrovski, A., 2021. On the class overlap problem in imbalanced data classification. *Knowl.-Based Syst.* 212, 106631. <https://doi.org/10.1016/j.knsys.2020.106631>.
- Wang, H., Meng, Y., Xu, H., Guan, X., Liu, Y., Liu, M., Wu, Z., 2024. Prediction of flood risk levels of urban flooded points using machine learning with unbalanced data. *J. Hydrol.* 630 (130742). <https://doi.org/10.1016/j.jhydrol.2024.130742>.
- Wilks, D., 2011. *Statistical Methods in the Atmospheric Sciences*, Third ed. Academic Press. ISBN: 9780123850225.
- Zhu, R., Guo, Y., Xue, J.H., 2020. Adjusting the imbalance ratio by the dimensionality of imbalanced data. *Pattern Recogn. Lett.* 133, 217–223. <https://doi.org/10.1016/j.patrec.2020.03.004>.